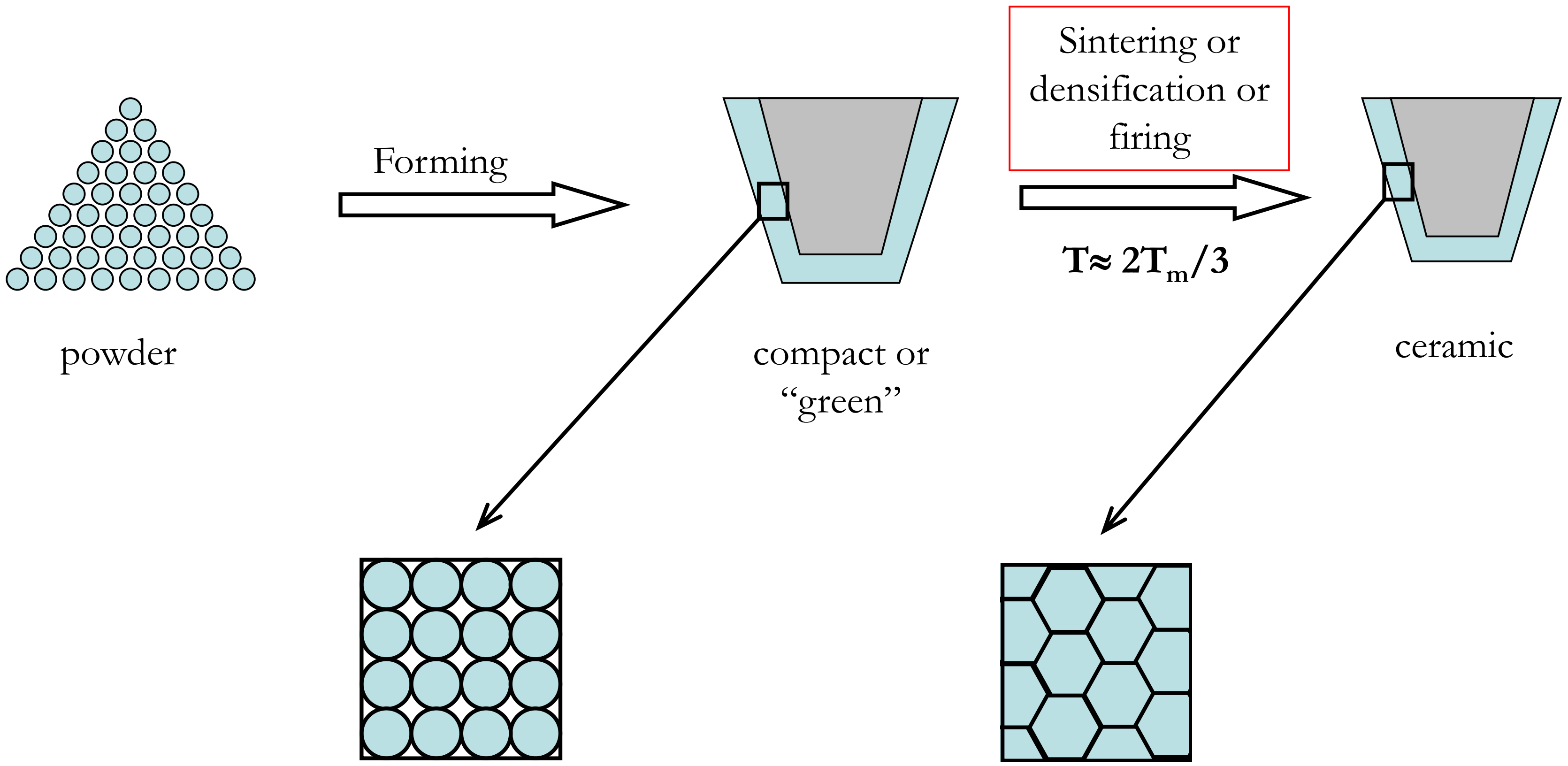
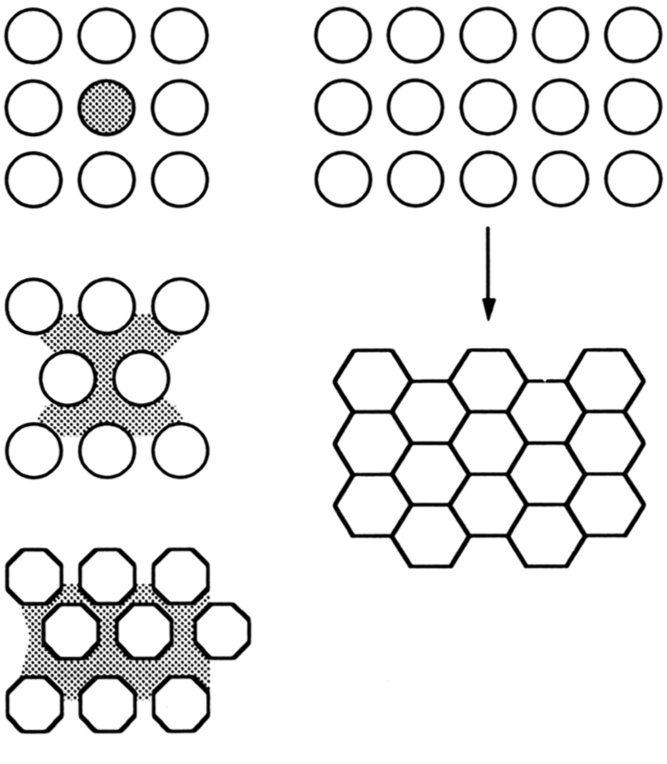
Chapter 7 Sintering and Grain Growth



Sintering

The starting material to form a ceramic material are powders, which are, by a variety of techniques, put into shape. The green body is thus a compacted, still porous body (30-60% of theoretical density) of grains, (idealized as spheres). This green body is subsequently heat treated. The usual sintering temperature is about 2/3 of the melting temperature. If there were such a thing as the holy grail of ceramic processing science, it probably would be how consistently to obtain theoretical density at the lowest possible temperature.



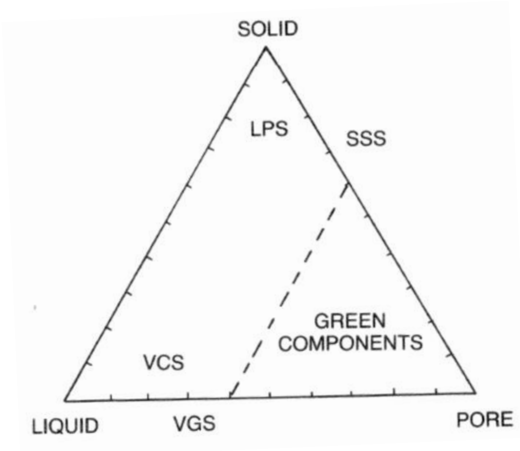
(*a*) (*b*)

(Barsoum, 1997)

Sintering is the process of forming a solid mass of material by heat and/or pressure without melting it to the point of liquefaction.

Types of sintering

Sintering: removal of pores between particles accompanied by shrinkage (densification) and grain growth.



Types of sintering

Solid-state sintering (SSS) : Conventional & Spark plasma sintering (SPS)
only in high-purity compounds

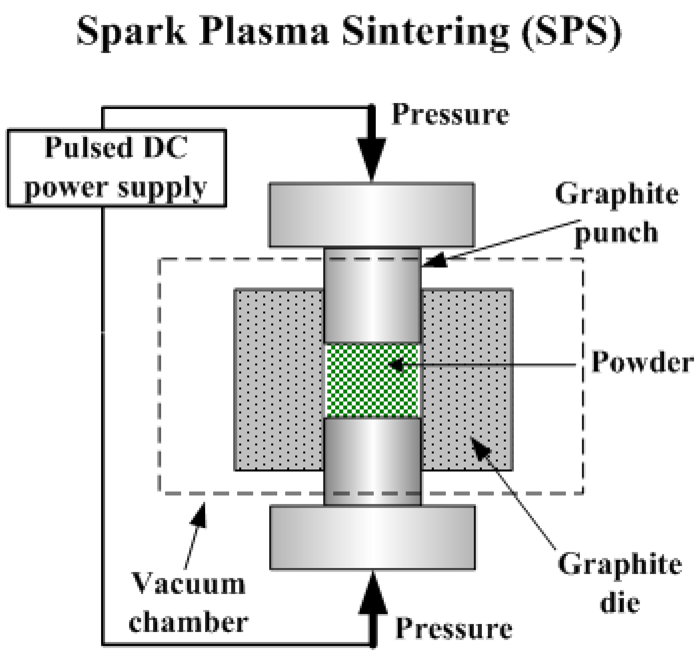
Liquid phase sintering (LPS)
<20% liquid; impurities or specific additives

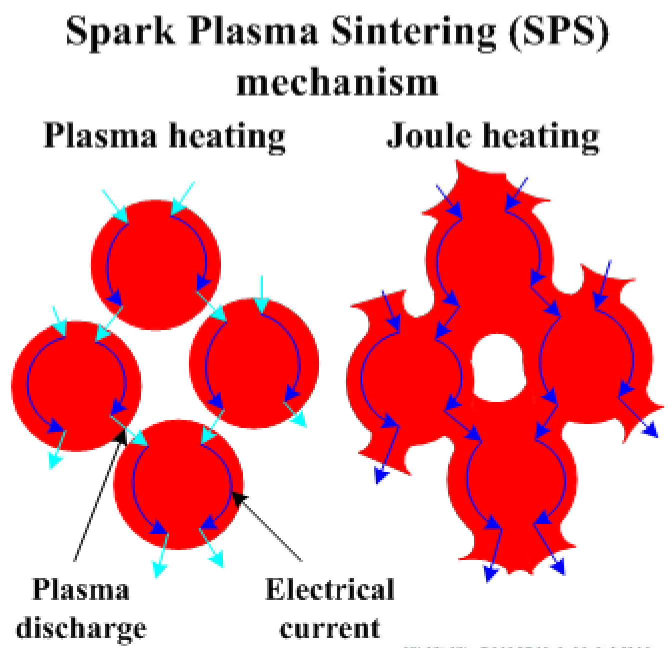
Viscous glass sintering or viscous flow (VGS)
Densification of glass powders

Viscous composite sintering or vitrification (VCS)
>20% liquid: whitewares, porcelains

- Sintering
  - solid state sintering
    - mono-phasic
    - poly-phasic
      - without reaction
      - with reaction
  - liquid phase sintering
    - liquid prop. <15%
    - liquid prop. >15%
  - pressure assisted sintering
    - hot pressing
    - hot isostatic pressing

Spark Plasma Sintering





The spark plasma sintering process proceeds through three stages:

Plasma heating: The electrical discharge between powder particles results in localized and momentary heating of the particles surfaces up to several thousands °C.

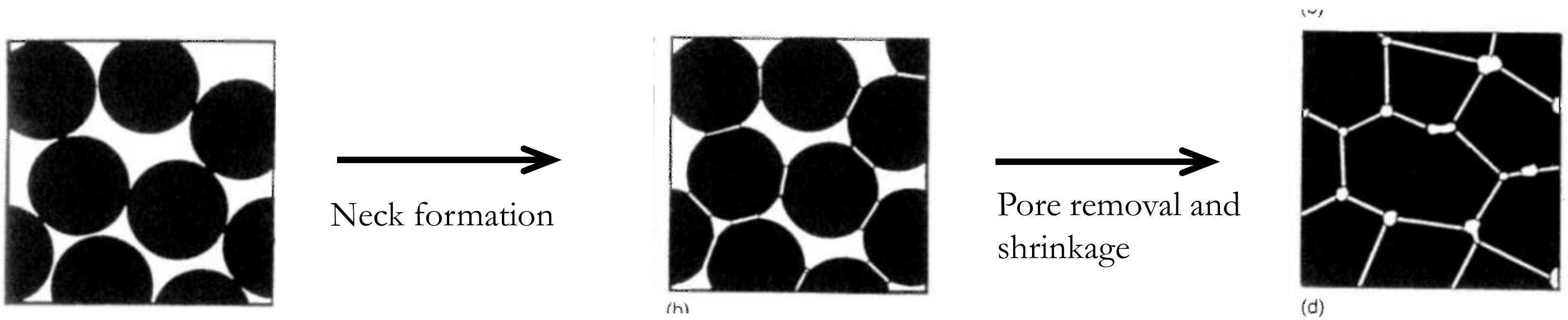
Joule heating: the joule heat increases the diffusion of the atoms/molecules in the necks enhancing their growth.

Plastic deformation: the heated material becomes softer and it exerts plastic deformation under the uniaxial force. Plastic deformation combined with diffusion result in the densification of the powder compact to over 99% of its theoretical density.

Driving force

Driving force for sintering: reduction of surface area and lowering of surface energy. High energy solid-gas surfaces are replaced by low energy solid-solid interfaces (grain boundaries).

At microscopic level, the driving force is related to the difference in surface curvature and consequently of partial pressure and chemical potential between different parts of the system.

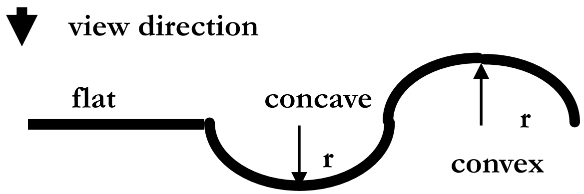


Effect of **particle size**: the smaller the particles, the higher the radius of curvature and the chemical potential → higher sintering rate.

Surface Energy I

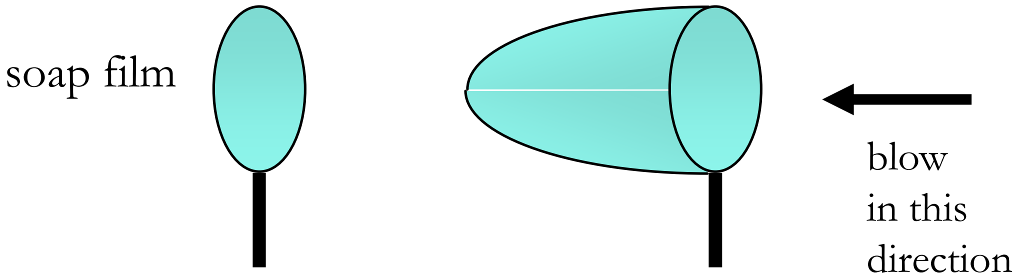
The macroscopic driving force operative during sintering is the reduction of the excess energy associated with surfaces. This can happen by (1) reduction of the total surface area by an increase of the particle size and/or (2) by the elimination of solid/vapor interfaces and the creation of grain boundaries.

Surfaces can be characterized according to their curvature and the respective radius of curvature:



radius: infinite negative positive

Extending a surface or changing the radius of curvature of a surface requires work:

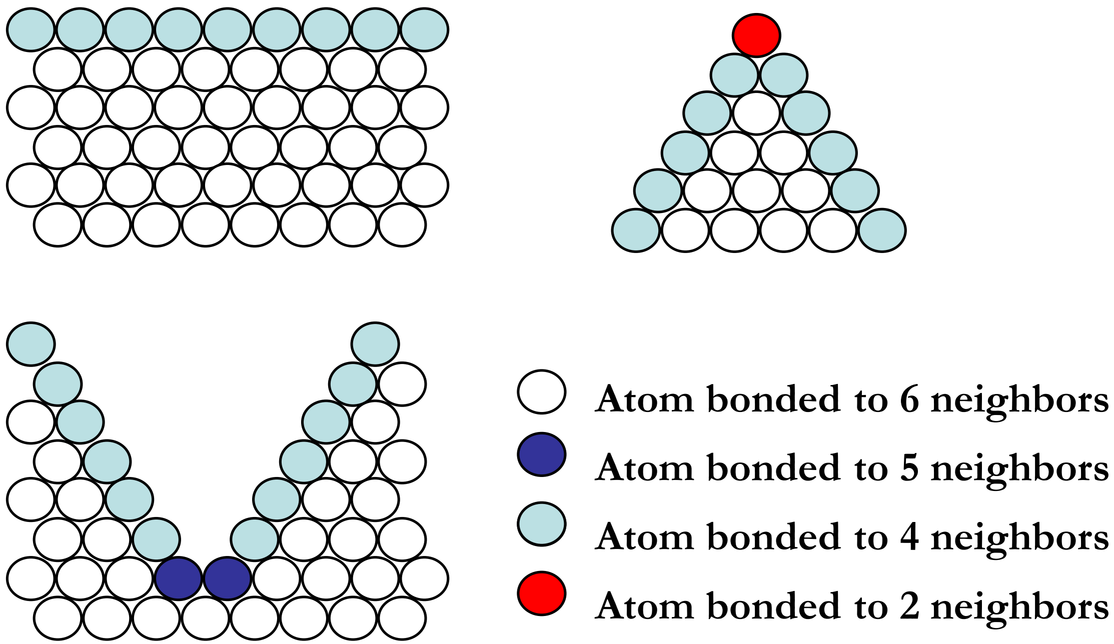


The surface energy γ is defined as the increase in free energy G per new surface A formed.

$$\gamma_E = \frac{\partial G}{\partial A} \quad (1)$$

Surface Energy II

The excess surface free energy is due to the different atomic arrangements along the surface relative to the bulk arrangement:



The average bonding of an atom is decrasing from concave over flat to convex surfaces, the partial pressure over the surfaces is increasing in the same order. The surface energy stored in small grains of ceramic materials is the driving force to coalesce (sinter)them together.

Surface Energy III

The work of expansion (mechanical work) of a bubble is equivalent to the increase in surface energy or

$$\Delta P dV = \gamma dA$$

The pressure difference necessary to bulge out a sphere by *dr* is given by:

$$\Delta P dV = \gamma dA \Rightarrow \Delta P = \gamma \frac{dA}{dV} = \gamma \frac{2}{r} \quad \textbf{(2)}$$

$$\frac{dA}{dV} = \frac{\left(4\pi r^2\right)' dr}{\left(\frac{4}{3}\pi r^3\right)' dr} = \frac{8\pi r dr}{4\pi r^2 dr} = \frac{2}{r} \quad \textbf{(3)}$$

The Gibbs free energy change due to this process is given by:

$$dG = VdP - SdT \quad \textbf{(4)}$$

Integrating for isothermal conditions yields:

$$\Delta G = \int_{P1}^{P2} VdP = VP_2 - VP_2 = V\Delta P \quad \textbf{(5)}$$

$$\Delta P = \gamma \frac{2}{r} \qquad \Rightarrow \Delta G = V\gamma \frac{2}{r} \quad \textbf{(6)}$$

Surface Energy IV

The change in chemical potential per formula unit of a species under a flat and a curved surface is thus

$$\frac{\partial \Delta G}{\partial n} = \Delta\mu = \mu_{curve} - \mu_{flat} = V_m \gamma \frac{2}{r} \qquad \textbf{(7)}$$

Where V_m is the volume per formula unit and *n* the number of formula units. At equilibrium this difference in chemical potential in the solid translates to a chemical potential difference in the gas phase i.e. a difference in partial pressure

$$\mu = kT \ln p \Rightarrow \Delta\mu = \mu_{curve} - \mu_{flat} = kT\left(\ln P_{curve} - \ln P_{flat}\right) = kT \ln \frac{P_{curve}}{P_{flat}} \qquad \textbf{(8)}$$

$$P_{curve} = P_{flat} \exp\left(\frac{2\gamma V_M}{kTr}\right) \qquad \textbf{(9)}$$

The partial pressure above a convex surface is thus higher than above a flat surface, and lower above a concave surface.

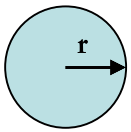
Similar considerations show that the curvature of a surface influences also the concentration of impurities and vacancies close to the surface. The concentration difference is given by

$$c_{curve} = c_{flat} \exp\left(-\frac{2\gamma V_M}{kTr}\right) \qquad \textbf{(10)}$$

The concentration under convex surfaces is therefore smaller than under concave surfaces.

Effect of curvature on thermodynamic properties

Laplace equation for a spherical droplet



$$\Delta P = \frac{2\gamma}{r}$$

Pressure difference across a curved interface. For a planar surface, $\Delta P = 0$

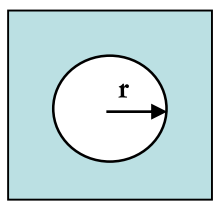
r (micron)	*P/P(r=∞)*
1	1.002
0.1	1.02
0.01	1.21
0.001	7.03

Effect of curvature on vapour pressure (Thomson's equation)

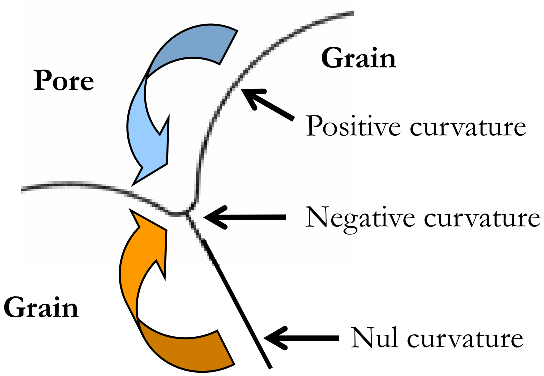
$$\ln \frac{P}{P(r=\infty)} = \frac{V}{RT}\left(\frac{2\gamma}{r}\right)$$

V: molar volume
γ: surface tension

If $r > 0$, $P > P(r=\infty)$



For a cavity ($r < 0$), $P < P(r=\infty)$

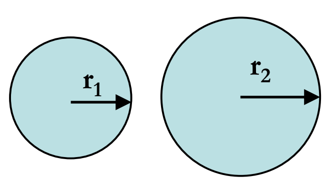


Effect of curvature on chemical potential

$$\mu_i - \mu_i(r=\infty) = \overline{V}_i\left(\frac{2\gamma}{r}\right)$$

Ostwald ripening

$$\mu_i(r_1) - \mu_i(r_2) = \overline{V}_i 2\gamma\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$



Particles with different curvature have different vapour pressure and chemical potential. Therefore they are not in equilibrium and the larger one will grow at the expense of the smaller one.

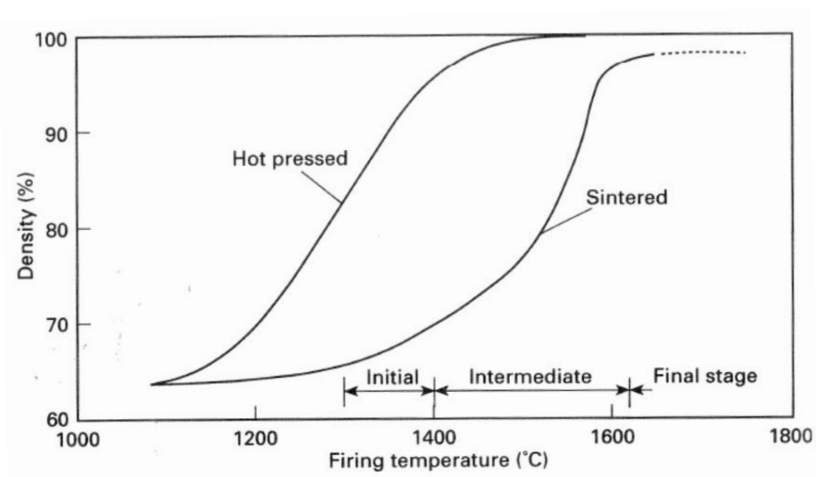
Stages of sintering

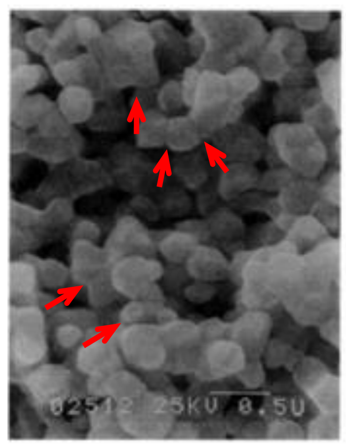
(a, b) Initial stage sintering. Formation of strong bonds and necks between particles at the contact points. Moderate decrease of porosity (initial 40-50%) from particle rearrangement.

(c) Intermediate stage sintering. The size of the necks increases and the amount of porosity decreases. The sample shrinks (the centers of the grains move towards each other. The grains transforms from spheres to truncated octahedra. This stage continues until pores are closed (r.d. 90%).

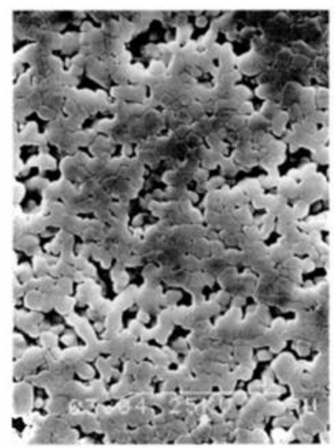
(d) Final stage sintering. Pores are slowly eliminated and major grain growth can occur.

In hot-pressing and hot isostatic pressing an additional driving force is provided by the external stress/pressure.

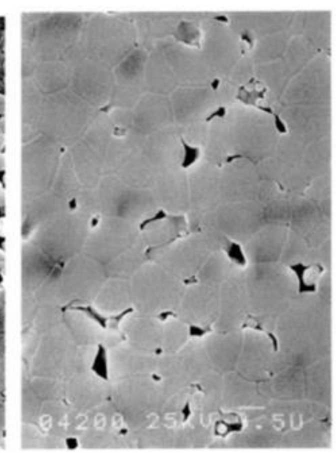




Initial stage

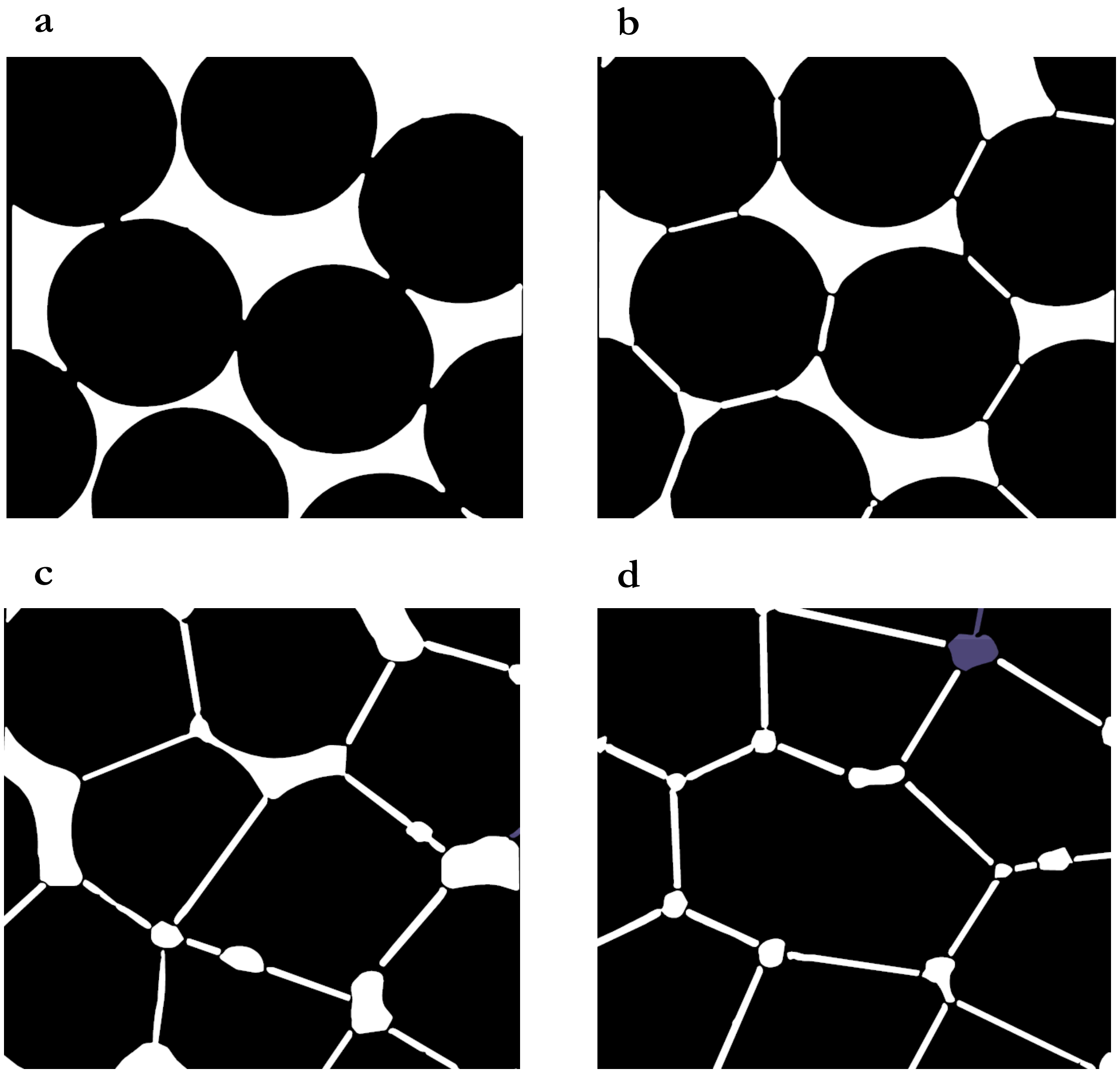


Intermediate stage



Final stage

Sintering stages I



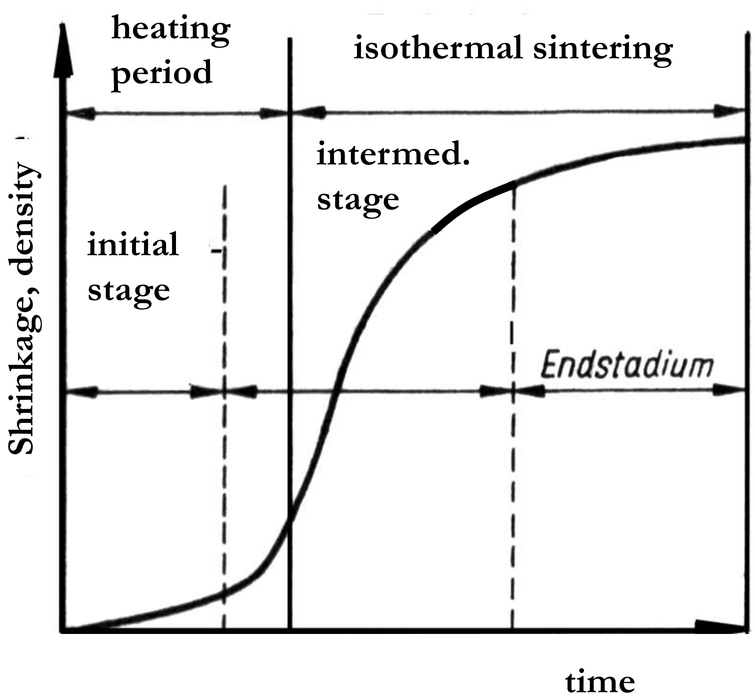
a) Green body, loose powder;

b) Initial stage: increase of the interparticle contact area from 0 to 0.2 grain diameter, increase of the density from 60 to 65%

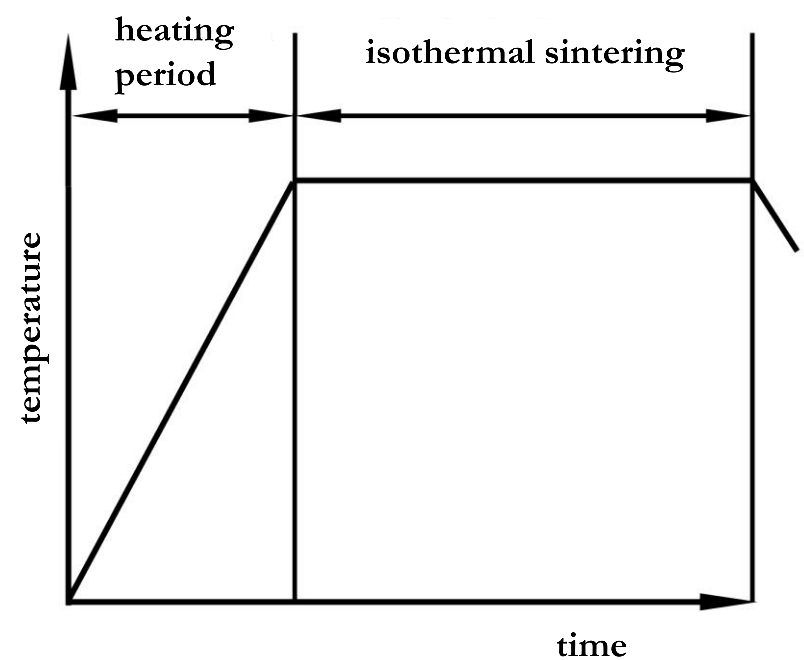
c) Intermediate stage: further increase of the contact area, stage characterized by continuous pore channels along three grain edges, increase of the density from 65 to 90%.

d) Elimination of the pore channel along three grain edges, increase of the density to 95 - 99%

Sintering stages II



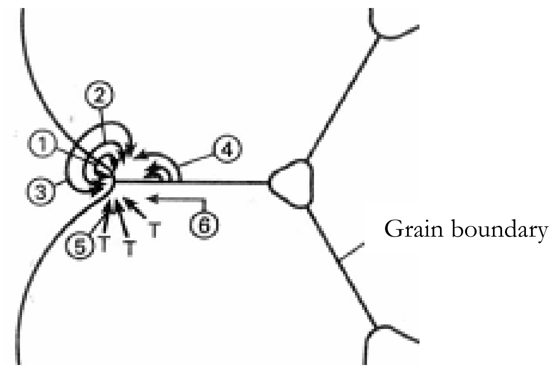
Development of density and shrinkage during a simple sintering cycle.



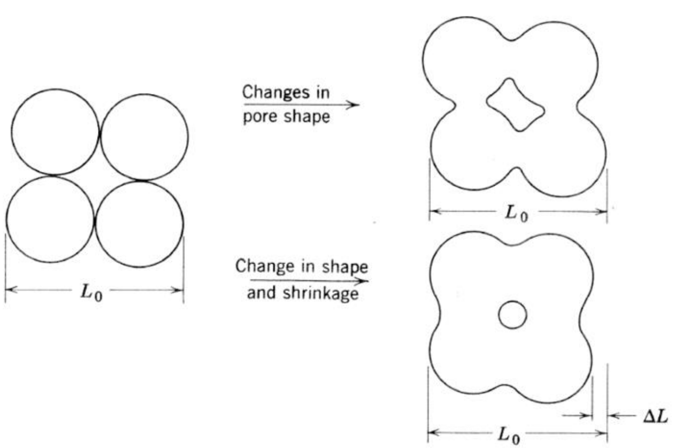
Temperature evolution during a simple sintering cycle.

Sintering Mechanisms

	Mechanism	Source	Sink	Densification
1	Surface diff.	Surface	Neck	No
2	Evaporation-condensation	Surface	Neck	No
4	Volume diff.	Grain boundary	Neck	Yes
6	Grain boundary diffusion	Grain boundary	Neck	Yes

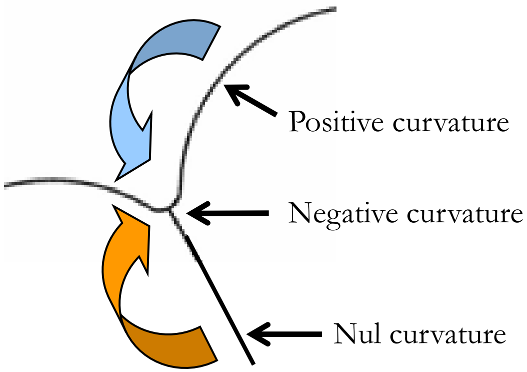


In ionic materials, **the mobility of the slowest moving species** dominates the diffusion process and sintering rates. This explain the strong dependence of sintering kinetics on nature and amount of uncontrolled impurities, dopants and sintering aids. Grain boundary diffusion is the most important densification mechanisms in many oxides.

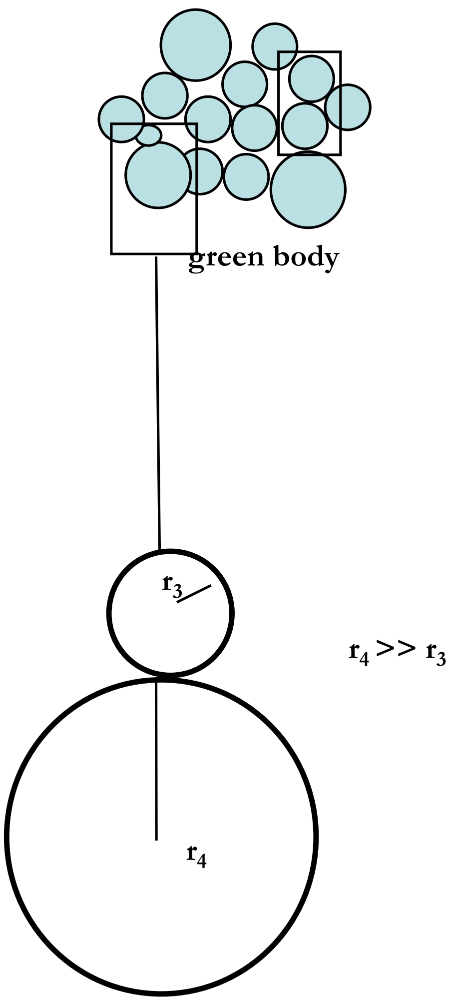


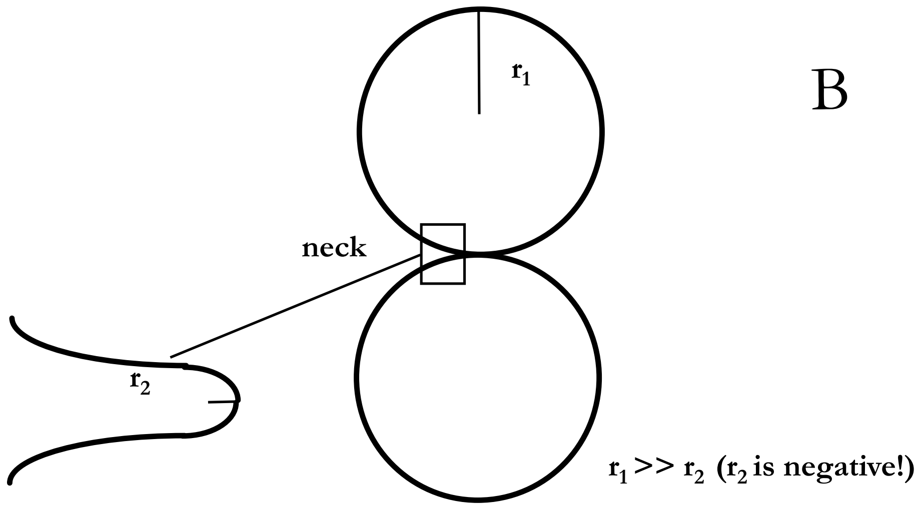
Surface diffusion & evaporation-condensation

Volume and grain boundary diffusion



Sintering mechanisms I



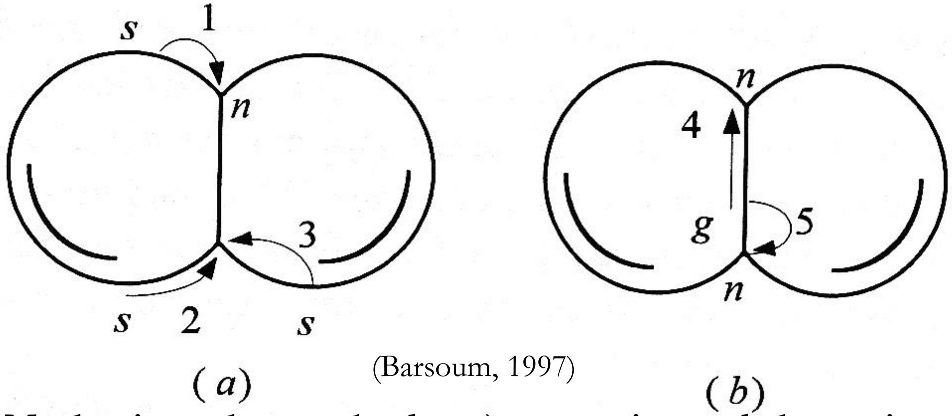


Due to the fine nature of the powders used as raw materials, the green bodies have a high internal surface area and, therefore, a high excess surface free energy. During the heat treatment the internal surface will be reduced and the necessary material movement is driven by the differences in surface curvature

A) between different sized grains, the big grains getting bigger, the small ones disappear = grain growth/shrinkage = overall coarsening of the texture

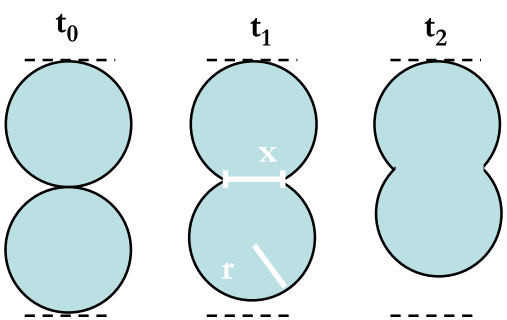
B) between the surface of two grains and the neck region between them = elimination of pore volume = densification

Sintering mechanisms II



(Barsoum, 1997)

Mechanisms that can lead to a) coarsening and change in pore shape and b) densification



Any mechanism in which the source of material is the surface of the particles and the sink is the neck area cannot lead to overall densification, because such a mechanism does not allow the particle centers to move closer together. For densification to occur, the source of the material has to be the grain boundary or region between powder particles, and the sink has to be the pore or the neck region.

Transport paths and mechanisms active during the sintering process:

1) diffusion through the gas phase in the porespace towards the neck area, evaporation - condensation

2) diffusion along the surface solid - gas towards the neck area

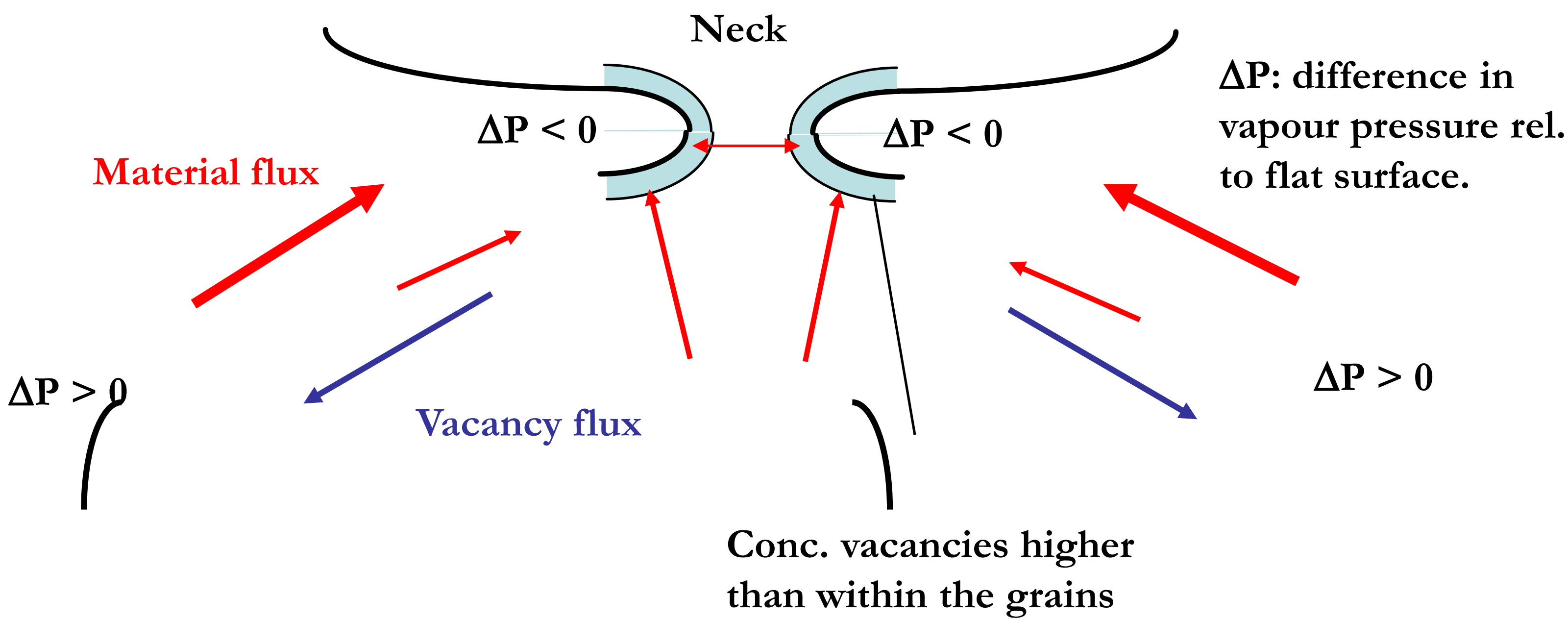
3) volume diffusion from the surface to the neck area

4) grain boundary diffusion from the the interface between the necks to the neck

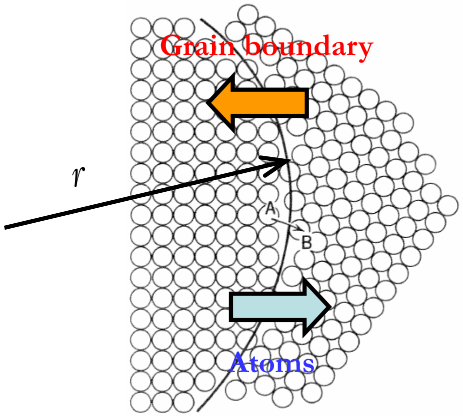
5) Viscous flow of material from area of high stress to areas of low stress

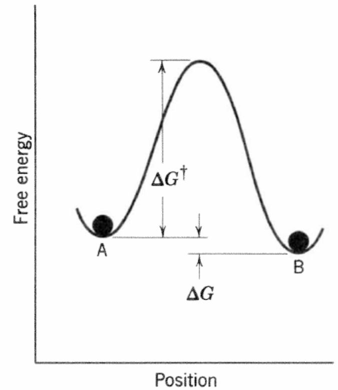
Sintering mechanisms III

Material moves to the neck regions of a polycrystalline solid because of the vapour pressure differences between the convex grain surfaces//grain boundary at the contact and the concave shaped necks.



Grain growth





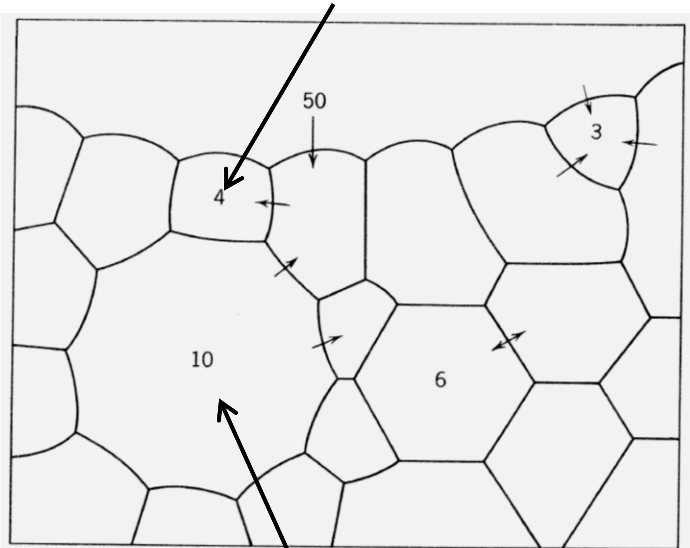
Driving force for grain growth: difference of chemical potential (Gibbs' free energy) across a curved interface

$$\Delta G = \overline{V}_i \frac{\gamma_{gb}}{r}$$

The grain boundaries with mobility M_{gb} migrate towards their centre of curvature at a velocity

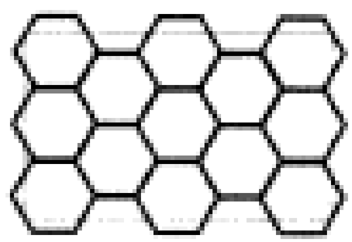
$$v = M_{gb} \gamma_{gb} \frac{1}{r}$$

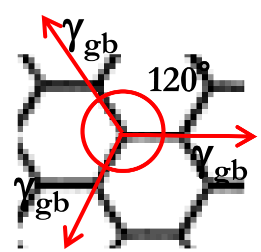
Concave boundaries

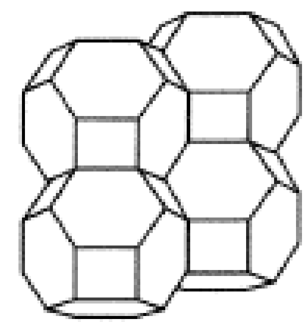


Convex boundaries

Grains with 6 sides: no grain boundary migration
Grains with <6 sides: the grains grow smaller
Grains with >6 sides: the grains grow larger

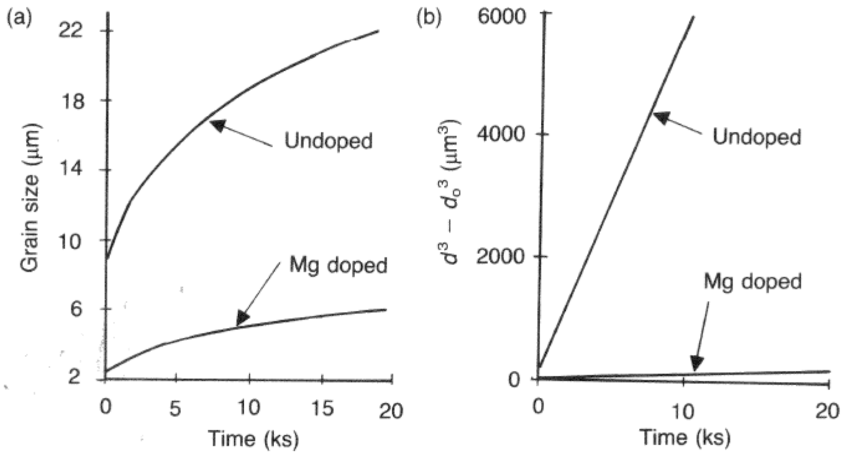






Factors affecting grain growth

General relationship: $d^m - d_0{}^m = kt$ m = 2-3; m = 2 if $\frac{d(d)}{dt} = \frac{k}{d}$



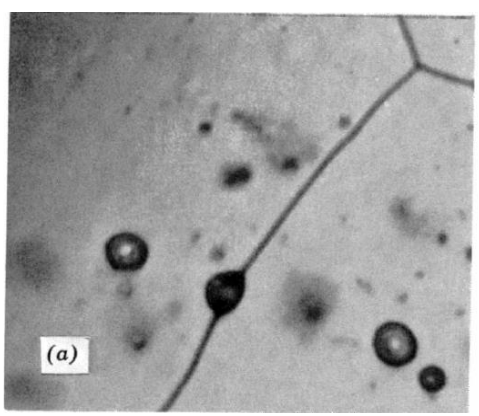
Grain growth in undoped and Mg-doped alumina

Dopants in solid solution affect depress grain growth because they segregate at grain boundaries reducing:
- the interfacial energy
- the grain boundary mobility

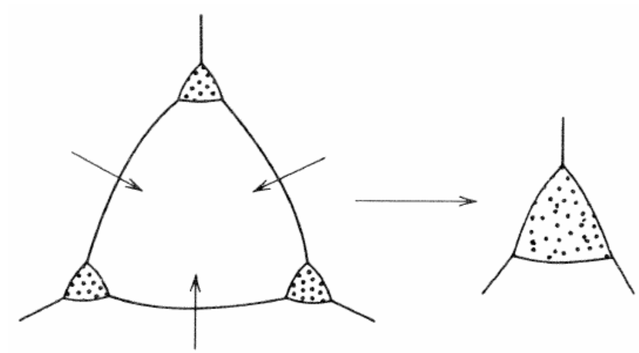
Grain growth is inhibited by pores, second phase inclusions and solid solution impurities. Pores and solid inclusions act as pinning centres for weakly curved grains. The critical grain size at which grain growth stops is given by (Zener):

$$D_f \approx \frac{d_i}{V_i}$$

D_f: limiting grain size
d_i: diameter of inclusions
V_i: volume fraction of inclusions



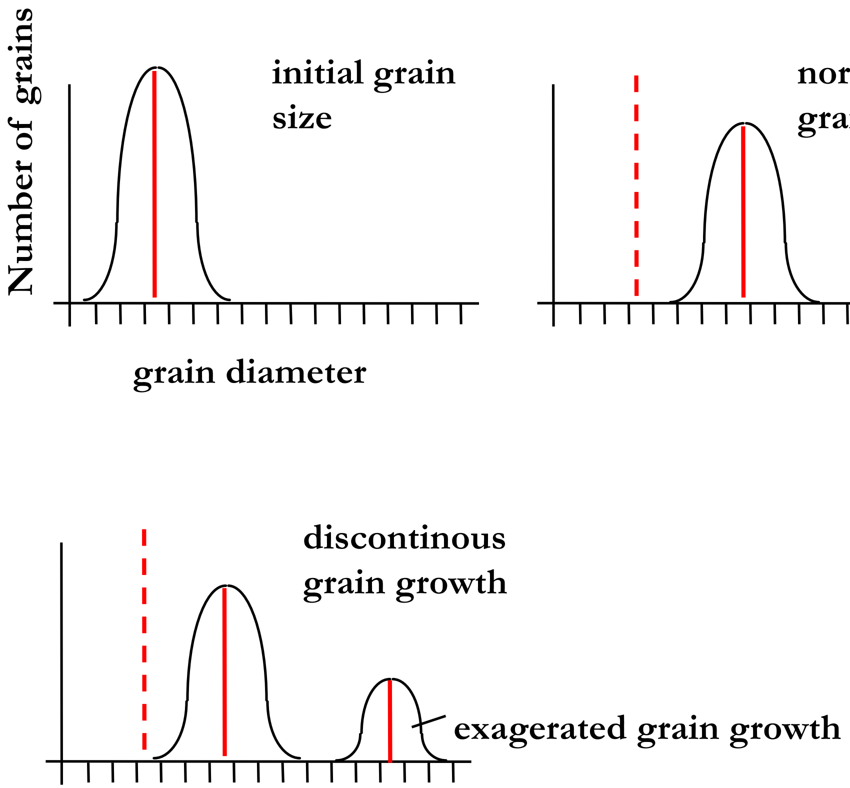
Grain boundary pinned by a pore

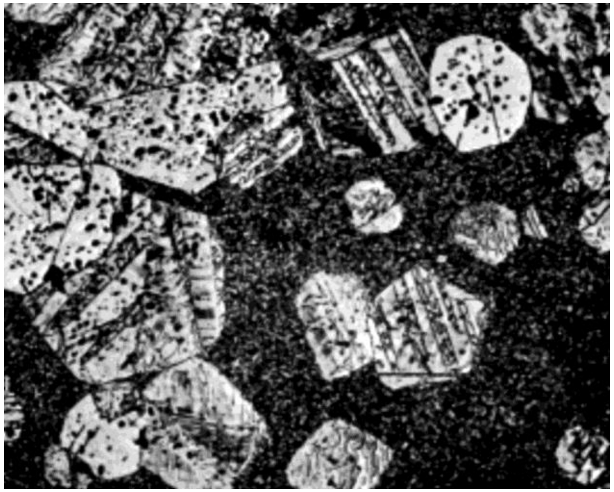


Dragging and agglomeration of pores determined by grain boundary migration

Grain Growth I

A process competing with sintering (= neck growth) is grain growth. The vapour pressure over small grains is higher than over large grains. There is, therefore, a net material flow from the little to the large grains, the small grains litteraly "evaporate". Two types of grain growth can be distinguished:





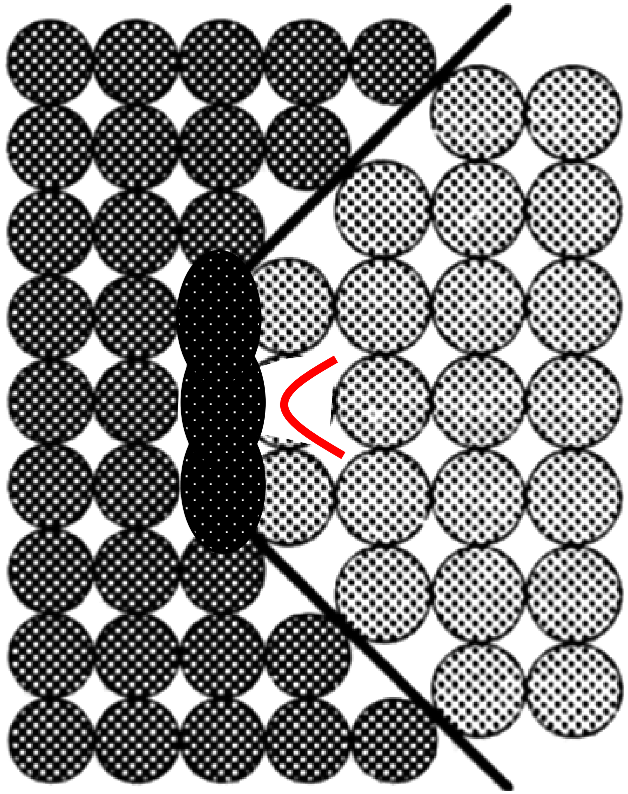
Bariumtitanate with extreme exagerated grain growth (250x)

Grain growth II

The driving force for grain growth is the dependence of the chemical potential of a species close to a surface from the curvature of the latter e.g.

$$\Delta\mu = \mu_{curve} - \mu_{flat} = V_m \gamma \frac{2}{r}$$

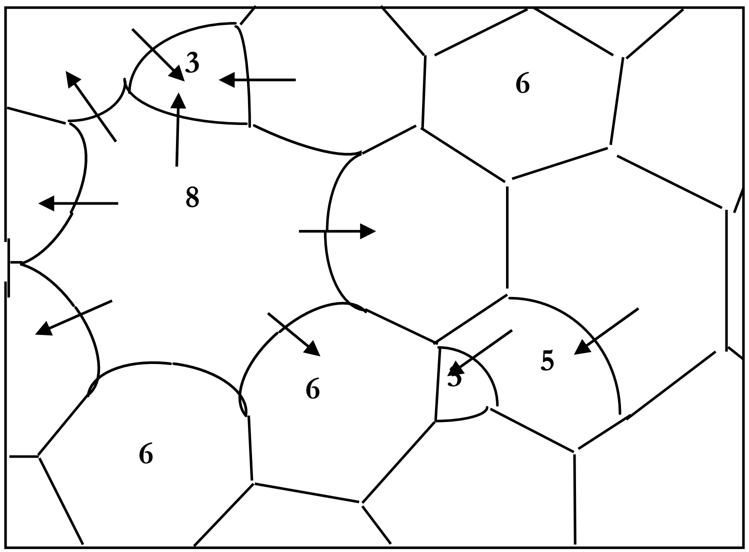
The situation on the right side shows a grain boundary between to grains with the same composition (shading difference is only to distinguish the grains). The atoms of the brighter crystal will transit to the darker crystal, because the latter has a concave surface e.g. a negative curvature radius *r.* Assuming the same surface tension γ, only a transfer from right to left will lead to a decrease in chemical potential e.g. free energy. The boundary will thus migrate to the right.



(Barsoum, 1997)

Equilibrium microstructure

A hypotetical pure 2-D ceramic material at equilibrium would have only six sided grains with faces joining under 120° . All faces would be straight. Grains with more than six sides would have convex faces and would grow, whereas grains with less than six sides would have concave faces and shrink.



Arrows indicate direction of movement；
Large becomes larger; to minimize grain boundary；
Grains with less than 6 sides, have convex boundary. Tend to shrink

The overall rate of grain growth depends on the boundary mobility *M*, which can be controlled by dopants, the surface energy of the moving boundary γ and the grain radius (r_0 : initial radius):

$$\frac{dr}{dt} = \frac{M\gamma}{r} \qquad r^2 - r_0^2 = 2M\gamma t$$

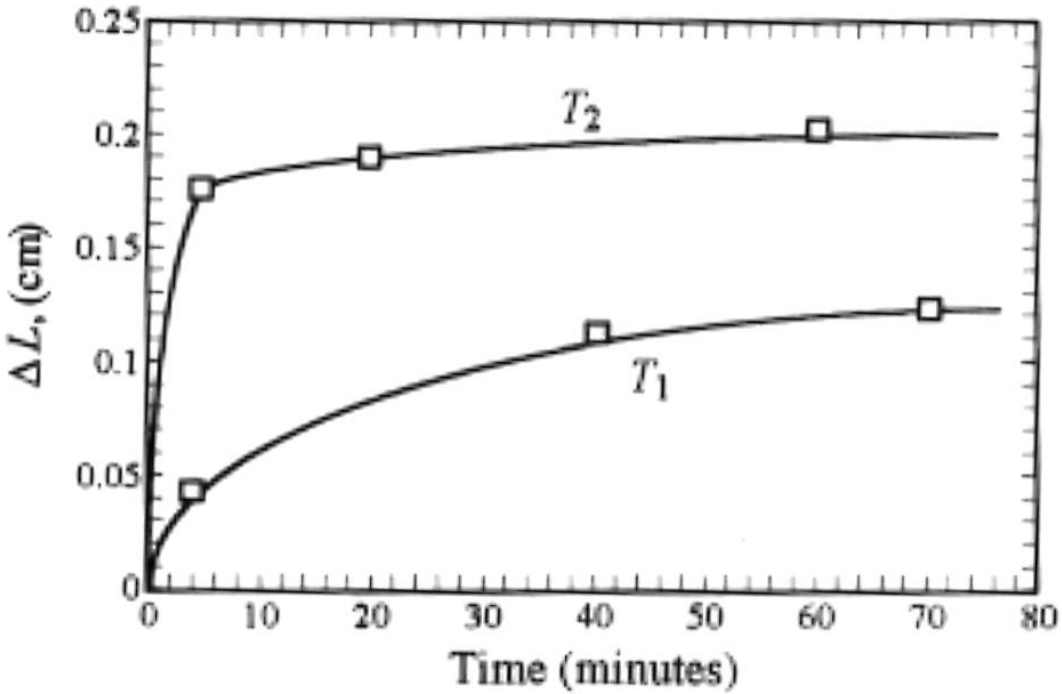
Sintering Kinetics

The mechanisms active at each sintering stage are different and have to be described by separate kinetic model. The general form of the shrinkage (= densification) equation in the initial stage can be given as

$$\frac{\Delta L}{L} = \frac{const}{kT}\frac{t^n}{r^m}$$

r: particle radius m: 1-1.5
t time (n: 1/3 -2/5)
L: length of the sintered body

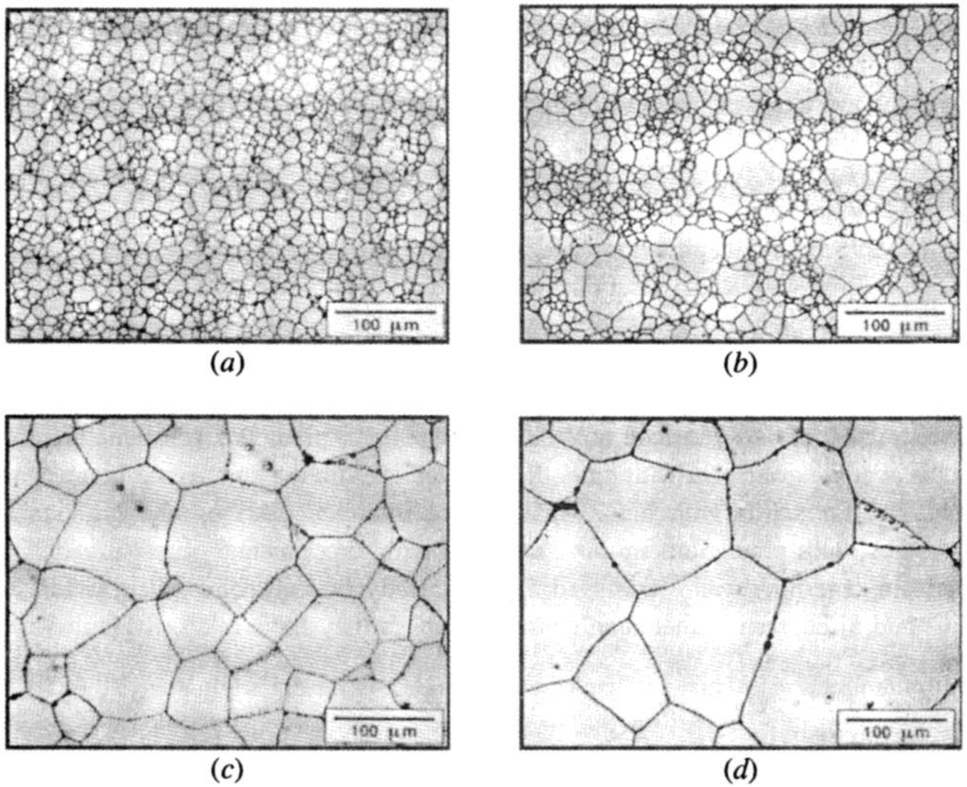
It is obvious from the above equation that ceramics with small grain size will shrink e.g. densify much faster than coarse green bodies.



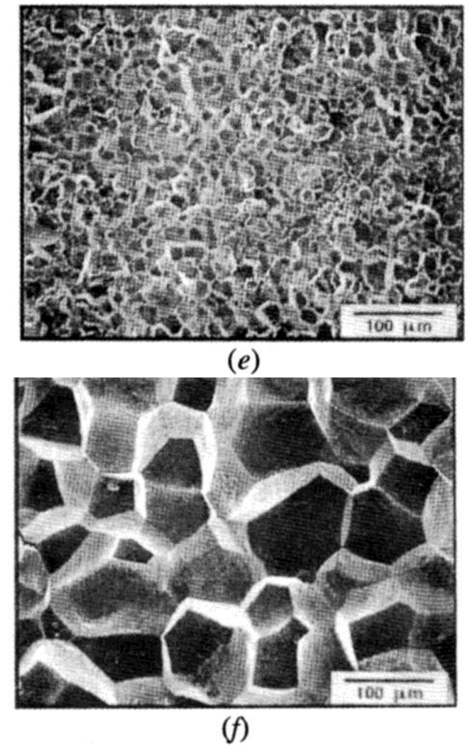
Typical axial shrinkage curve as function of temperature $T_2 > T_1$.

(Barsoum, 1997)

CdI Microstructural Evolution



Grain growth evolution in a hot pressed CdI pellet sintered at 103MpA and 100°C for a) 5, b) 20 c) 60 and d) 120 min. Observe the discontinuous grain distribution after 20 min.



Fracture surfaces of the samples shon in (a) and (d)

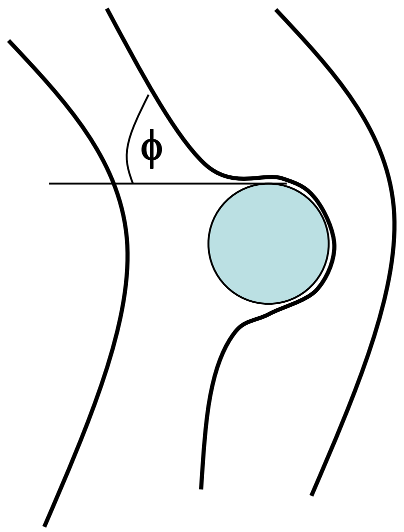
(Barsoum, 1997)

Grain Boundary Pinning

Impurities or dopants are often concentrated at interfaces. When a grain boundary moves, e.g. during grain growth, the solutes concentrated at the interface have to be carried along. The diffusion coefficient of the doping elements such as MgO in Al_2O_3, $CaCl_2$ in KCl, ThO_2 in Y_2O_3 or Nb in $BaTiO_3$ are smaller than the diffusion coefficient of the main constituents of the ceramic, thus slowing down the grain boundary mobility called solute drag.

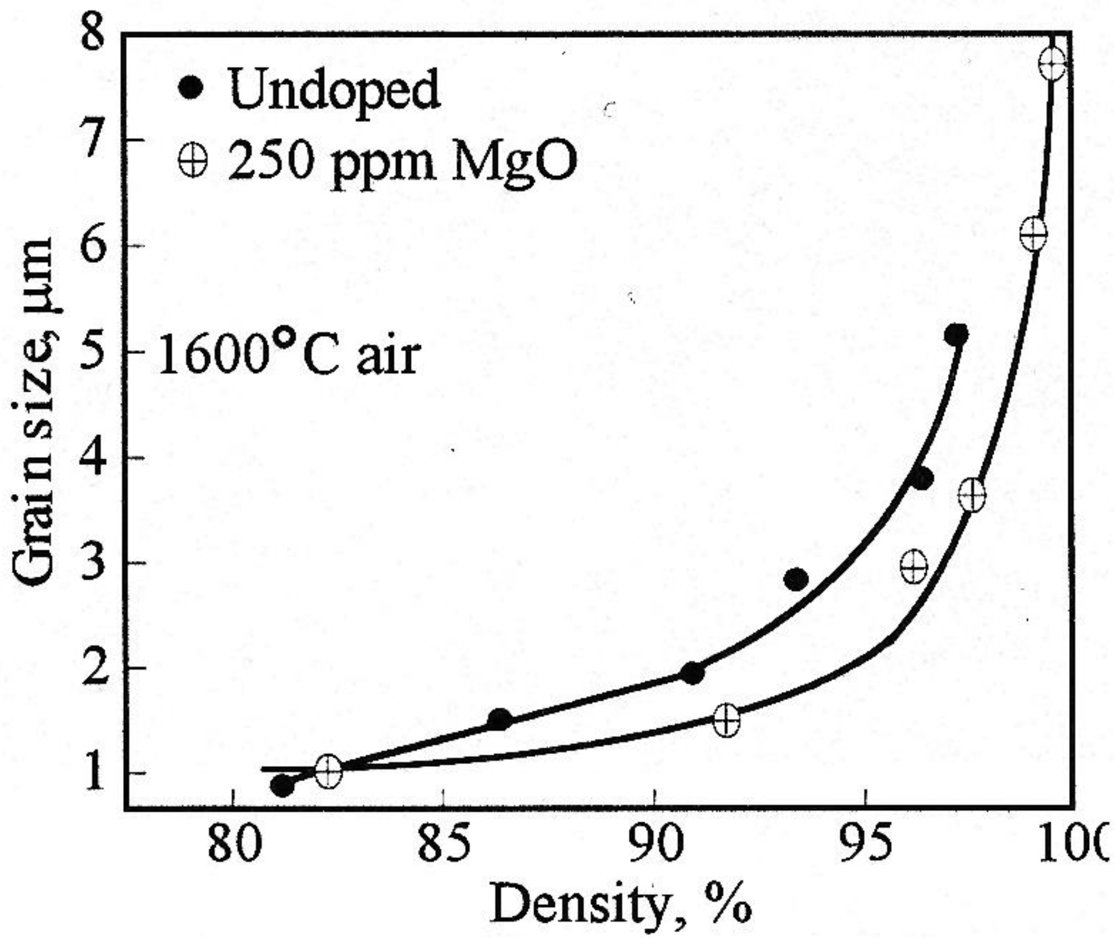
Pores and second phase inclusions have the same effect. The retaining force is a function of the inclusion radius *r* and is given by

$$F = 2\pi\gamma r \sin\phi \cos\phi$$



Increase of interface area while passing an inclusion

Doping and Grain Boundary Pinning



Grain size and density evolution of an alumina ceramic with and without MgO doping.

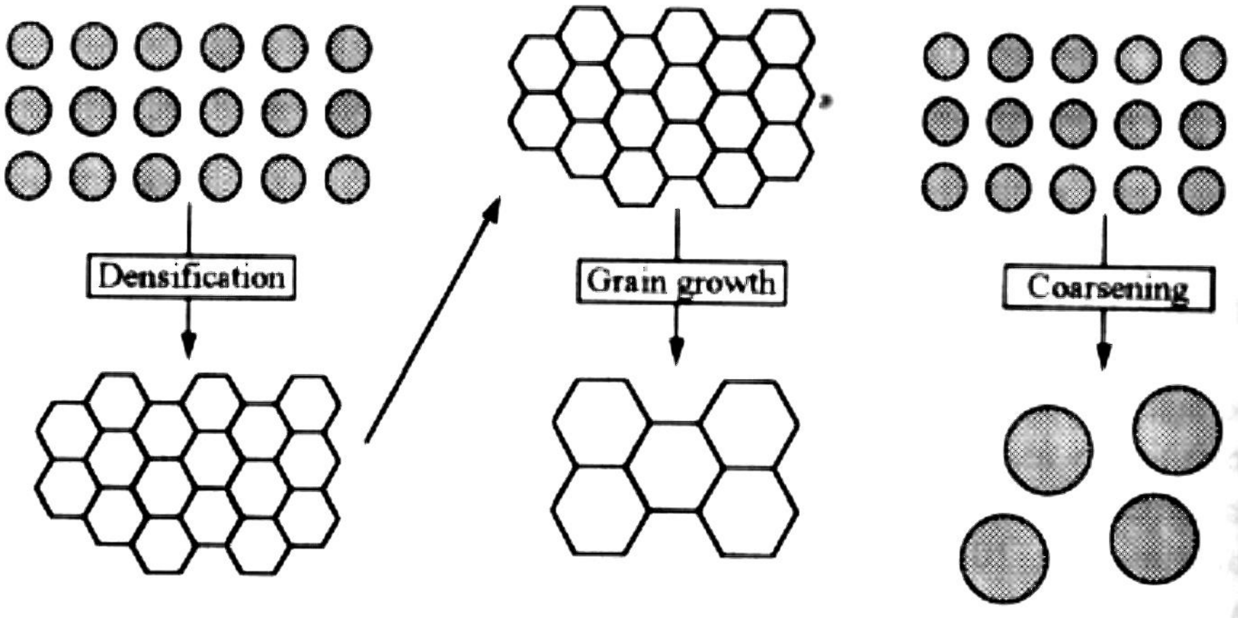
(Barsoum, 1997)

Densification vs. Coarsening I

Densification and grain growth occur simultaneously. The resulting textures and density depend on which mechanism is predominant.

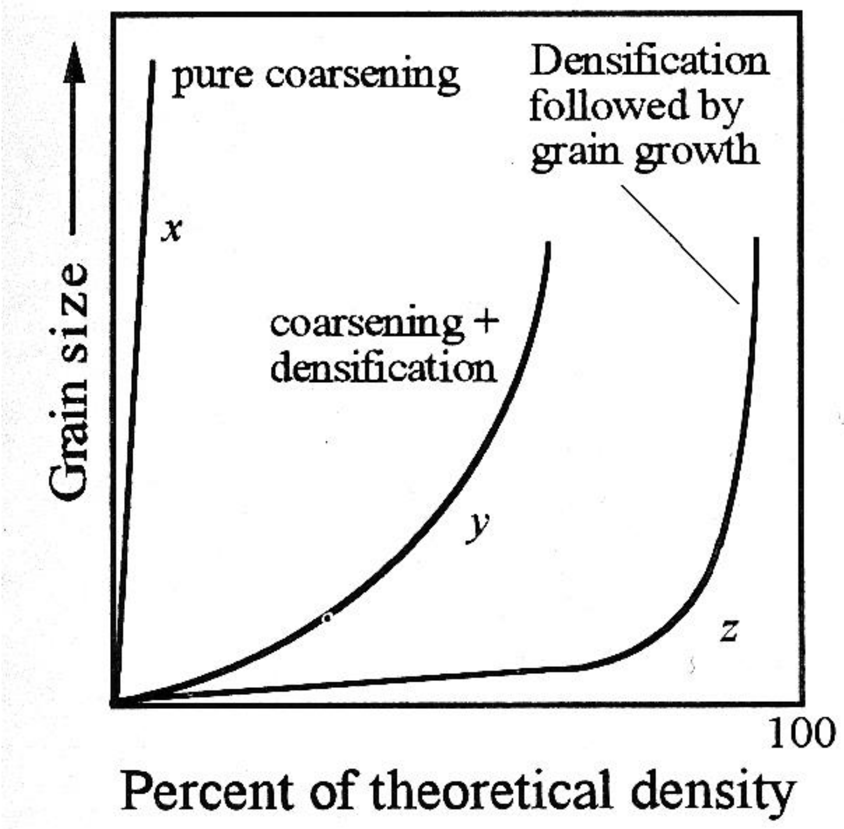
a) Densification followed by grain growth

b) Coarsening alone



(Barsoum, 1997)

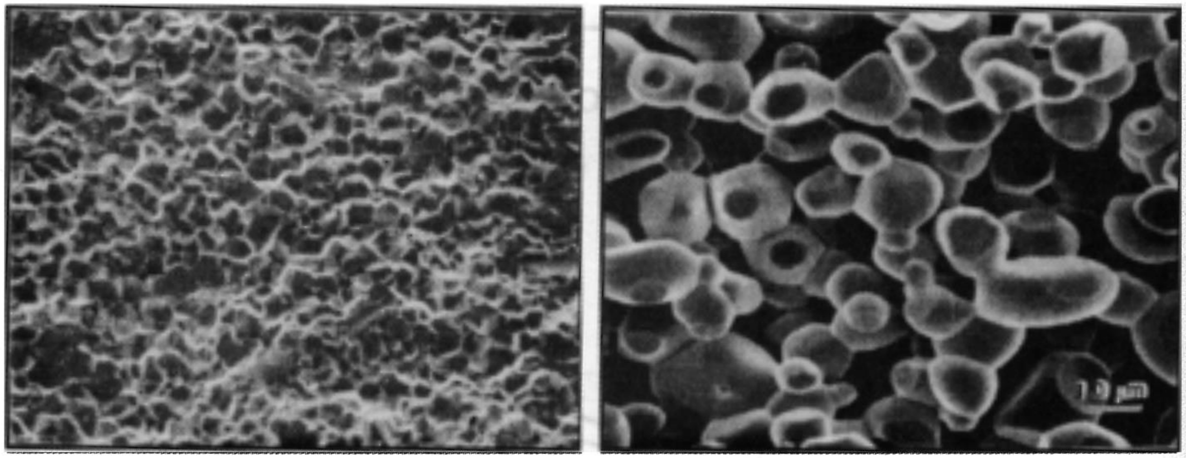
Densification vs. Coarsening II



Only densification followed by grain growth will give good final densities.

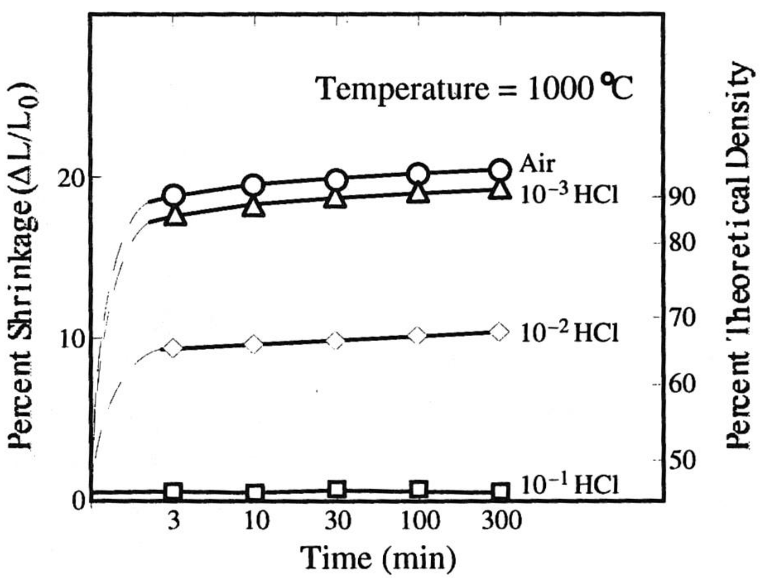
(Barsoum, 1997)

Densification vs. Coarsening for Hematite



(a) (b)

Hematite sintered at 1000° in
a) air
b) an Argon/10%HCl

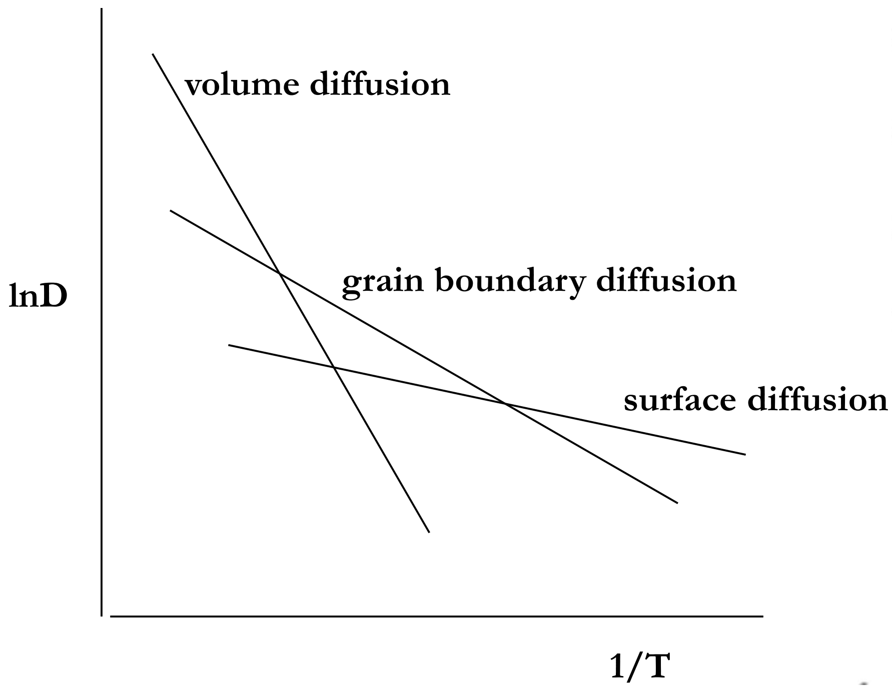


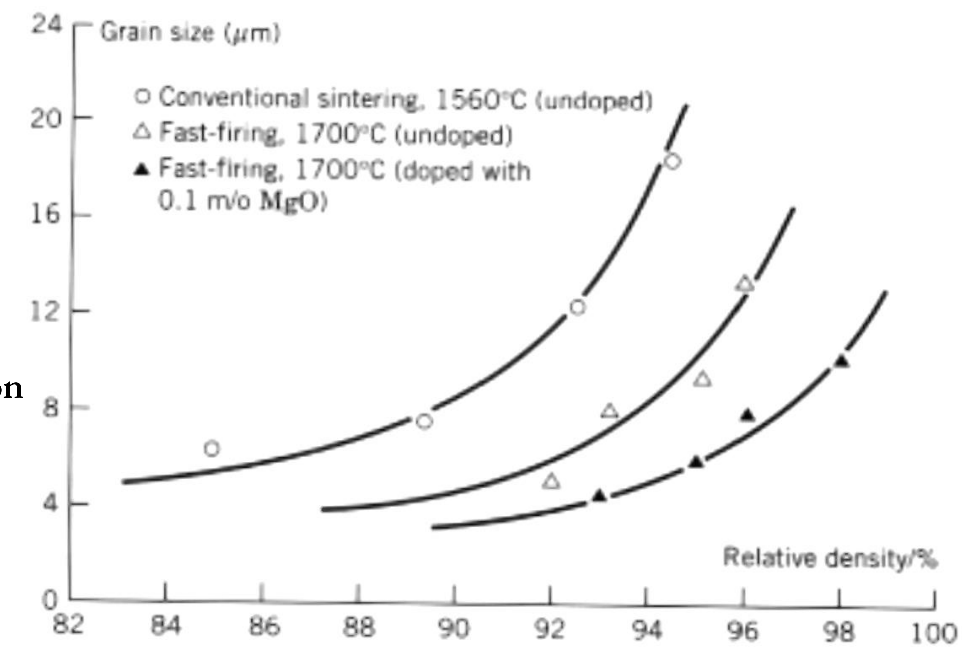
Shringkage and density curves for hematite sintered in different atmospheres. HCl in the sintering atmosphere enhances the grain coarsening mechanisms e.g. only little densification occur.

(Barsoum, 1997)

Suppressing Coarsening

Fast firing and doping of the starting material are ways to suppress grain growth





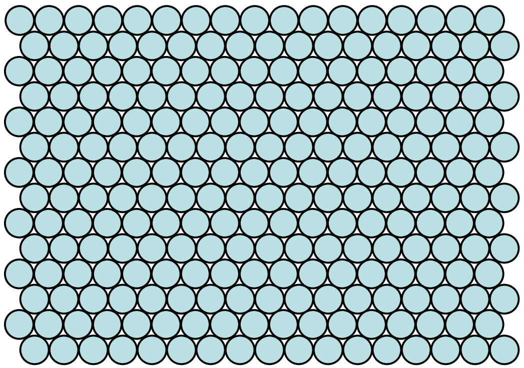
Surface diffusion from one grain to the next is the most effective mechanism for grain growth. Surface diffusion is the fastest process at low temperature. With a fast heating rate at the beginning the sample is only a short time in the low temperature domain.

Difference in densities and grain sizes in fast fired and and conventionally sintered alumina

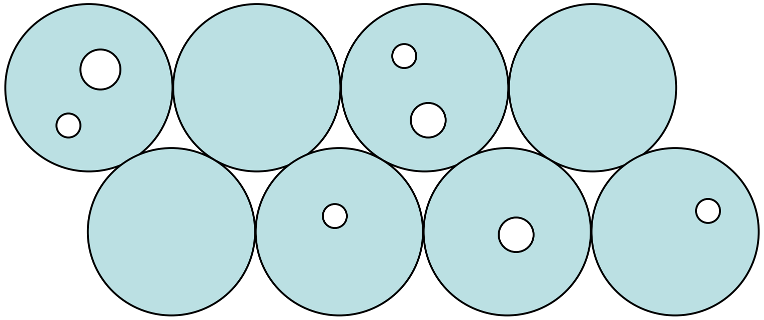
(Barsoum, 1997)

Porosity of sintered bodies

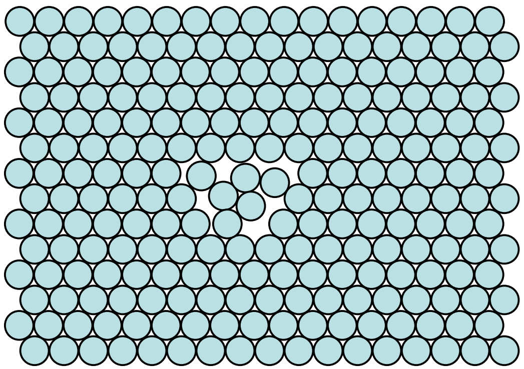
Pores are important and unwanted elements of a ceramic microstructure. The final pore space in an sintered ceramic is mainly a function of the pore volume in the greenbody. Source of pores



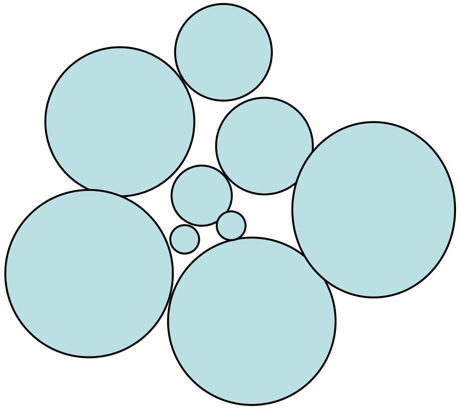
Intergranular pore space. Ordered packing of monodispersed spherical particles minimizes initial pore volume.



Intragranular pore space



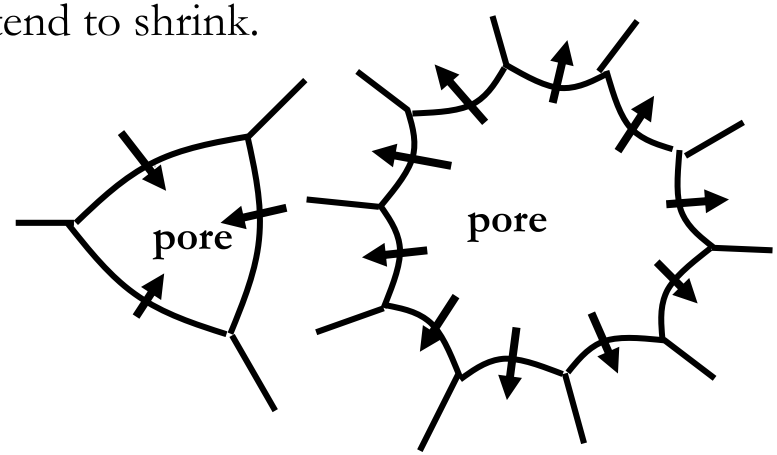
Extra pore space due to hard, disordered agglomerates



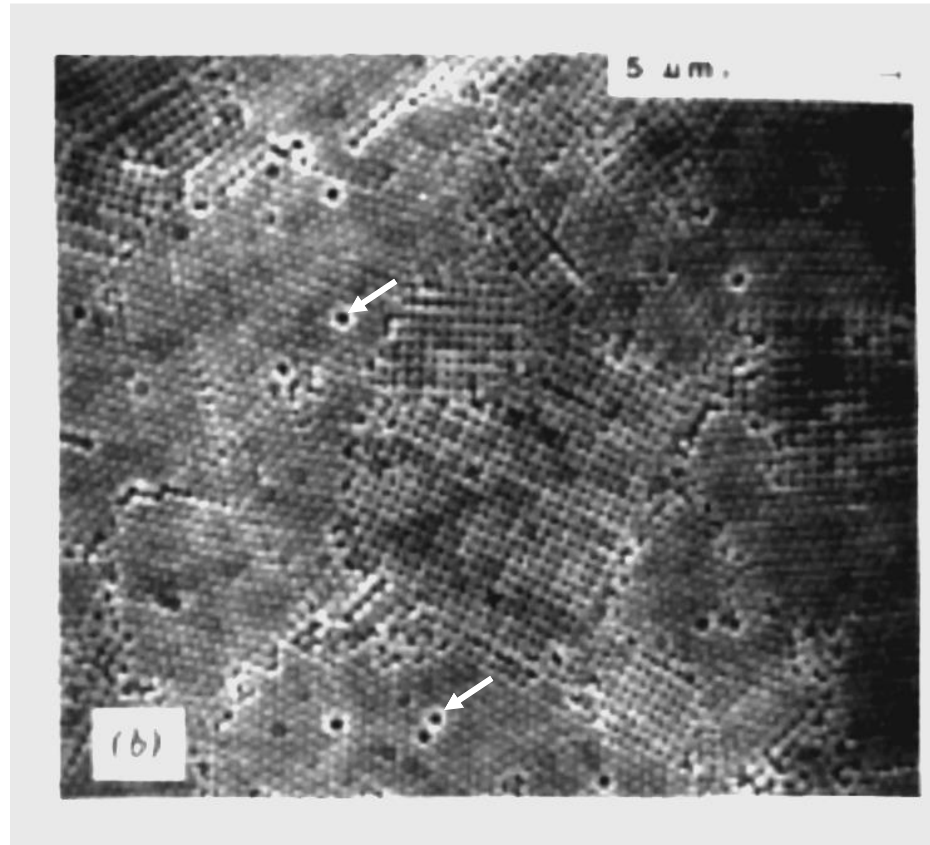
Extra pore space due to polydispersed powders.

Evolution of porosity I

Pores can like grains grow or shrink. The two parameters affecting pore growth is the number of surrounding grains and the dihedral angle. Generally pores with few neighboring grains tend to shrink.

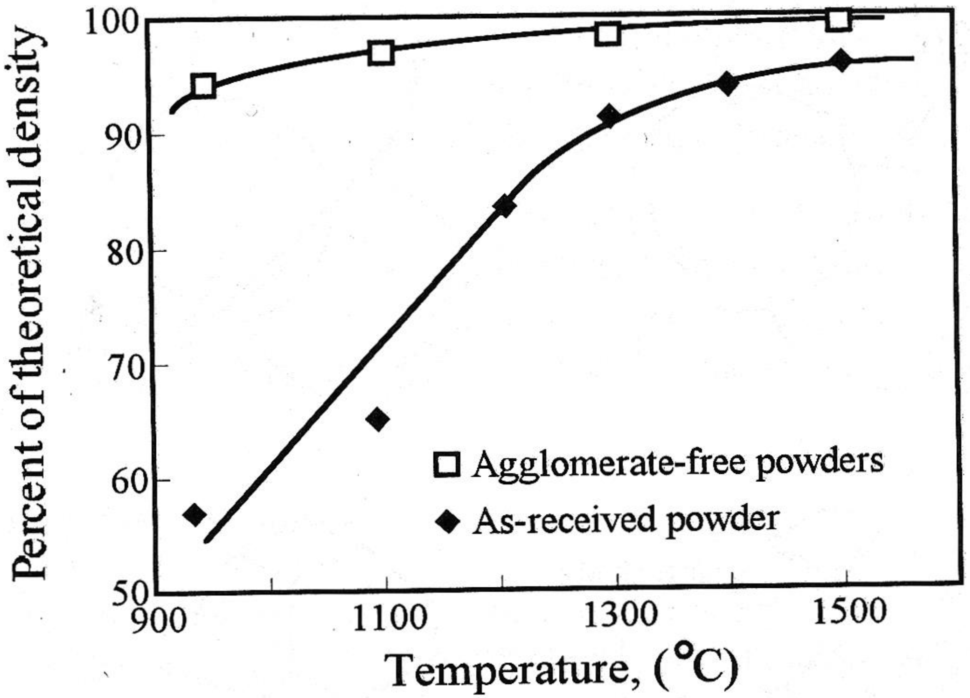


(Small) pores with few neighboring grains have concave surfaces and tend to shrink, whereas (large) pores with many neighboring grains have convex surfaces and tend to grow

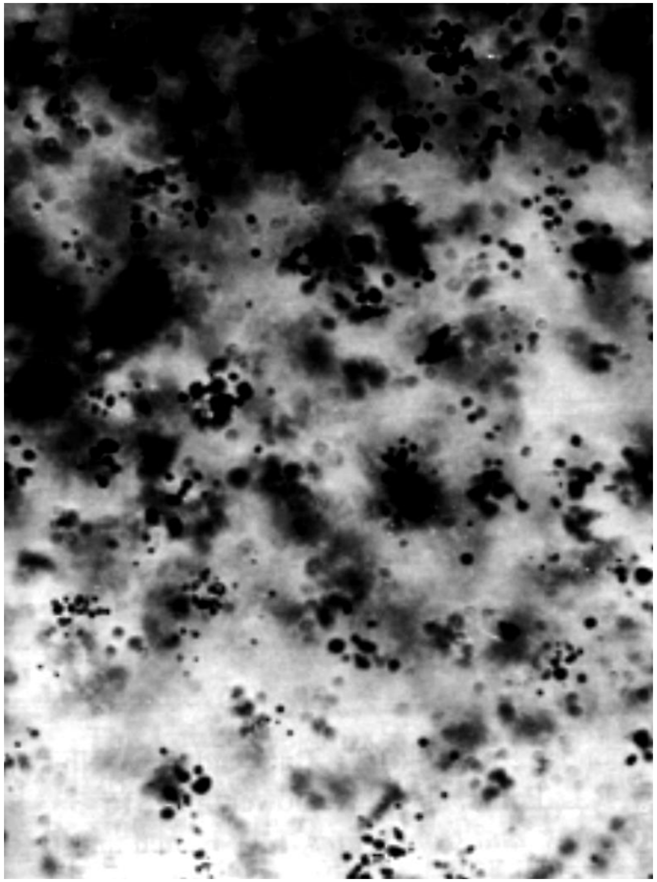


Monodisperse greenbody of boron doped SiO_2. The larger pores may be difficult to evacuate.

Evolution of porosity



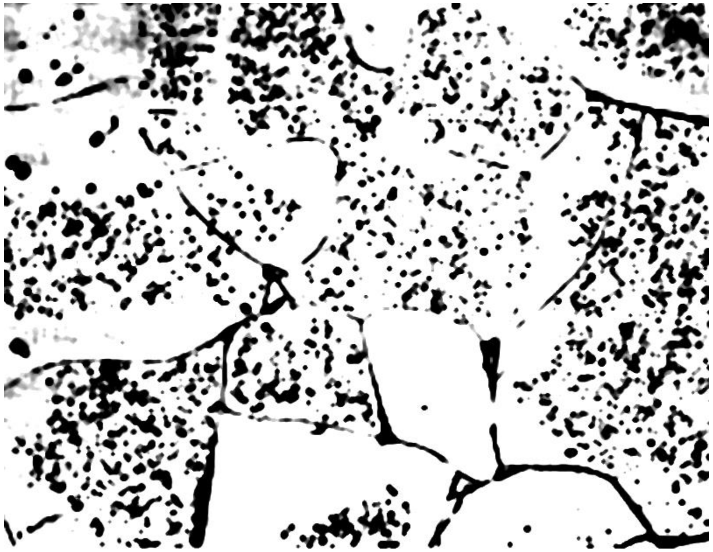
Sintered density of yttrium stabilized zirconia as function of powder processing. Eliminating agglomerates (through milling) enhances the sintering kinetics and allows to achieve higher densities.



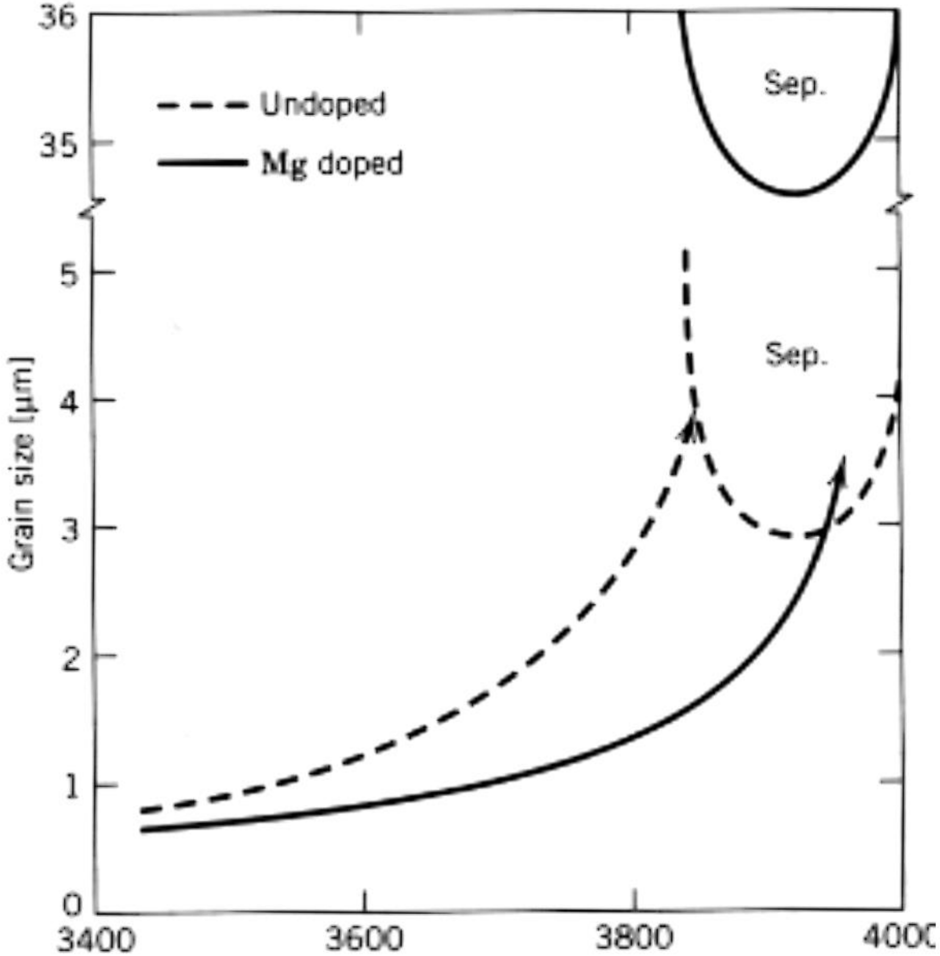
Residual pore clusterss resulting from improper powder processing (left over hard agglomerates) in a yttrium oxide ceramic doped with 10 mole % thorium oxide (transmitted light, 150x)

Pore entrapping

Internal pores can be a primary feature of the powder, but they can also be entrapped boundary pores. Entrapping happens when grain growth is to fast. Slowing down grain boundary mobility will also prevent entrapping.



Entrapped pores in the center of an undoped alumina ceramic (500x)

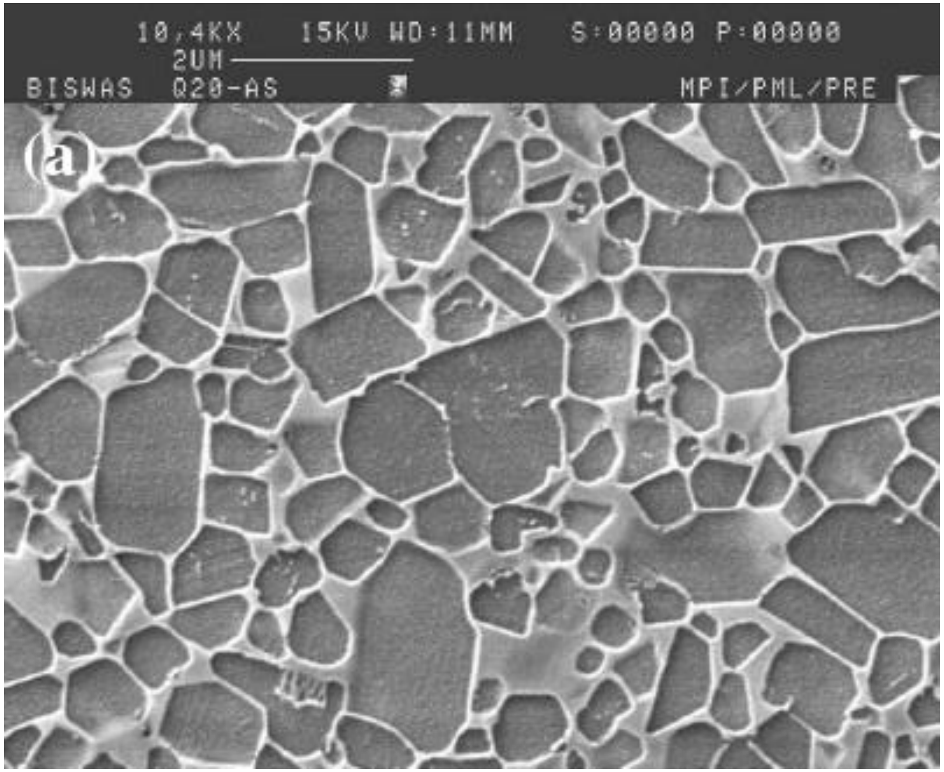


Density as function of grain growth for alumina. Adding MgO as doping will slow grain growth and displace the region where pore entrapping (=separation) occurs.

(Barsoum, 1997)

Liquid phase sintering

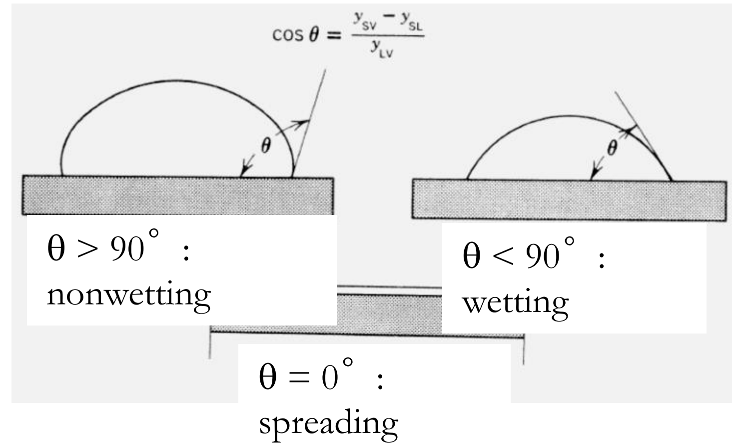
Second phases, that melt at the sintering temperature, are often present as impurities in the starting powder or a added on purpose. The small amount of liquid present during sintering will significantly alter the kinetics and the properties of the sintered body. Liquid phase sintering is important for the manufacturing of a large range of materials. It enhances the rate of sintering and the homogeneity of the final sinter body.

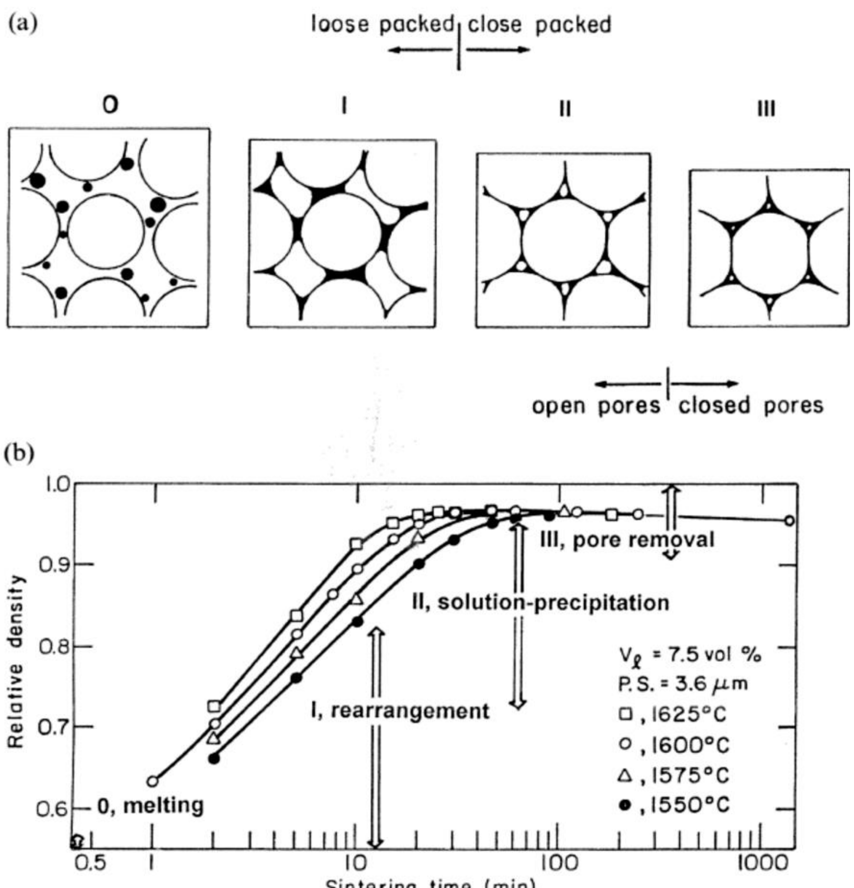


Microstructure of as-sintered SiC with gadolinium and holmium oxides. The oxides were liquid during the sintering and were quenched to a glass (Biswas, thesis, 2002)

Liquid phase sintering

• Enough liquid phase must be present (1-5 vol.%).
• The liquid must wet the solid (contact angle θ<<90°).
• The solid must be partially soluble in the liquid.
•Driving forces are higher for small particles (stronger capillary forces) with high surface energy and high solubility





Particle rearrangement: the liquid spread on the particles which rotate and slip. Significant densification occurs, up to 70%, without modification of particle and pore morphology.

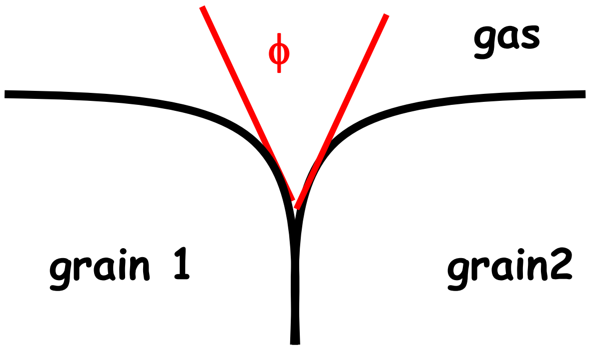
Solution-precipitation:
(1) Ostwald ripening. Small particles dissolve in the liquid and the material precipitates on bigger particles because solubility depends on the radius of curvature.
(2) Dissolution occurs in the neck region because of Laplacian compressive force and material redeposit away of the neck region.
(3) Sharp corners dissolve and material precipitates on regions of lower curvature.

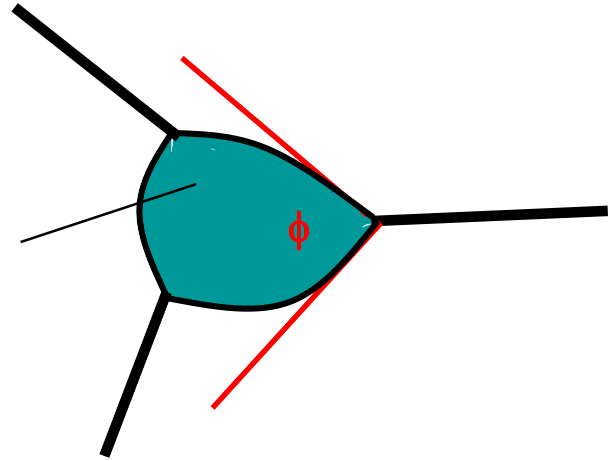
Coalescence. When enough grain growth has occurred, a solid skeleton is formed and the pores becomes closed (at ≈90% re. dens.). Pore elimination can proceed by solid-state diffusion.

Wetting

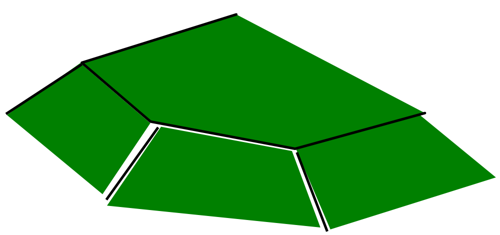
The behavior of the liquid depends on the solid - liquid surface tension. The expression for the dihedral angle in two phase systems with a liquid or a gas and a solid phase are:

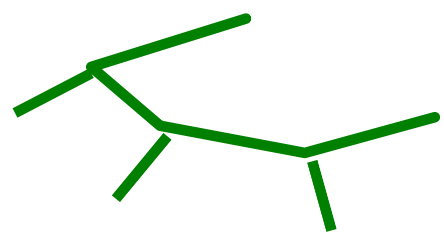
$$\gamma_{SS} = 2\gamma_{SG(L)} \cos\frac{\phi}{2}$$

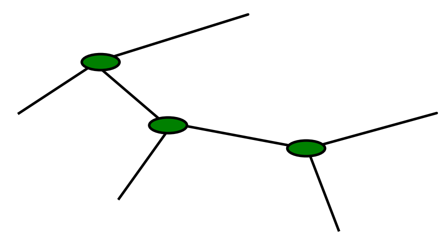




In a solid-liquid system the liquid will be distributed according to the dihedral angle:







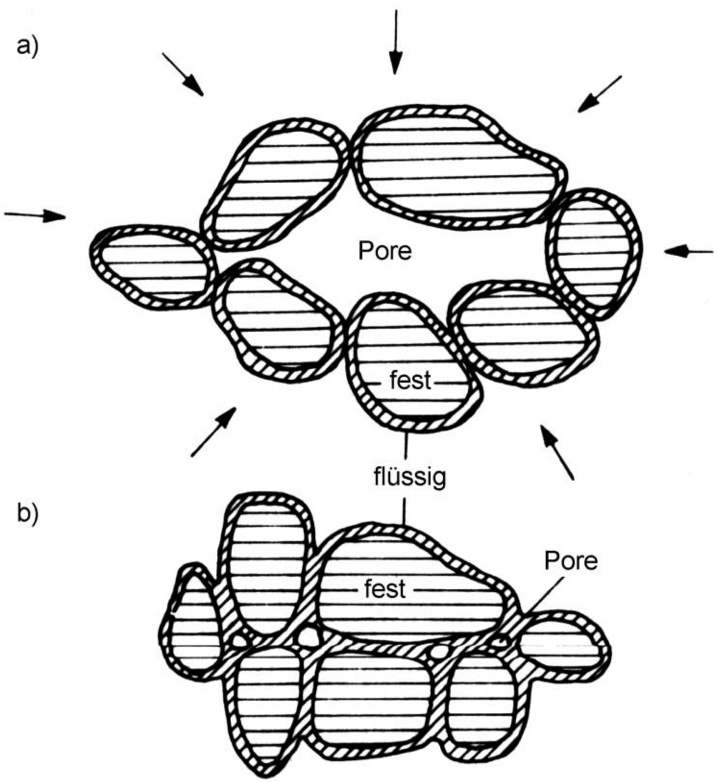
ϕ=0° all faces and edges are wetted (covered by a fluid film)

0°< ϕ<60° all edges and cornes are wetted.

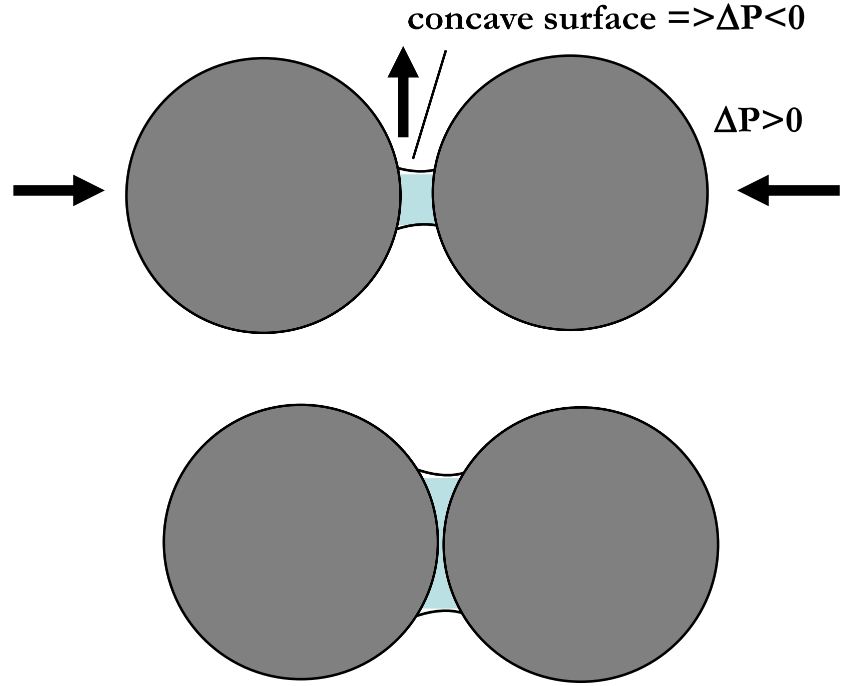
ϕ>60° only three or more grain junctions are wetted

Liquid assisted compaction

The fluid present will support the rearrangement of grains. For wetting angles $\theta < 45^{\circ}$ with the liquid distributed along the edges, the grains will be pulled towards the common edge due to the negative curvature of the liquid-solid surface and the associated pressure difference (capillary effect). This pressure difference will assist the compaction during sintering. Example: ZnO varistors are sintered with the addition of Bi_2O_3, which is only slightly soluble in ZnO and forms an eutectic liquid at the sintering temperature of ZnO.

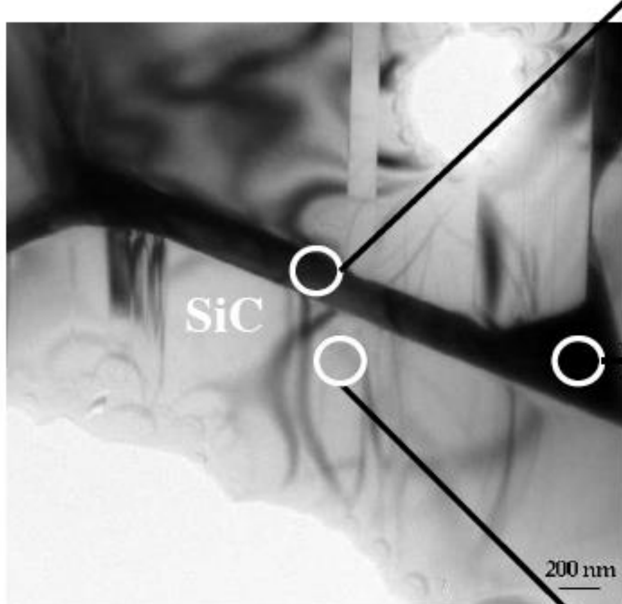


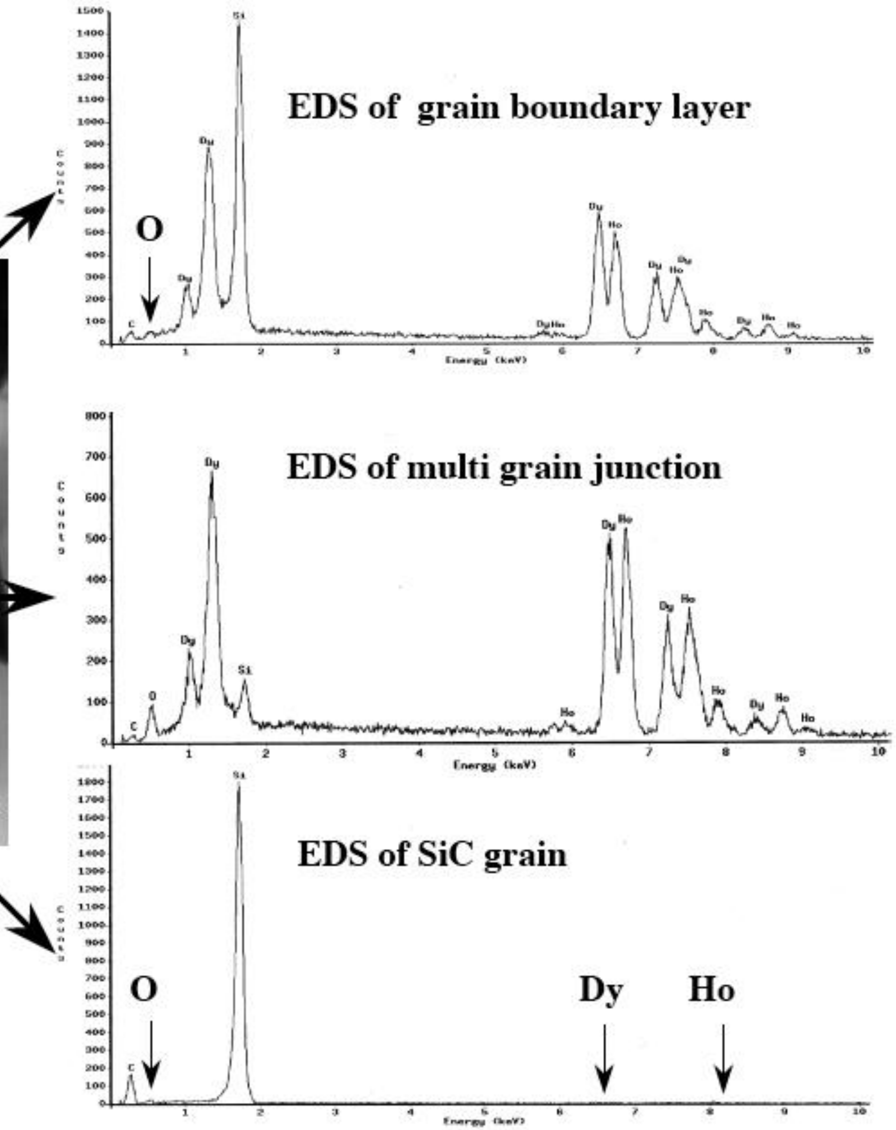
Grain rearrangement



Capillary action

Liquid phase sintering of SiC

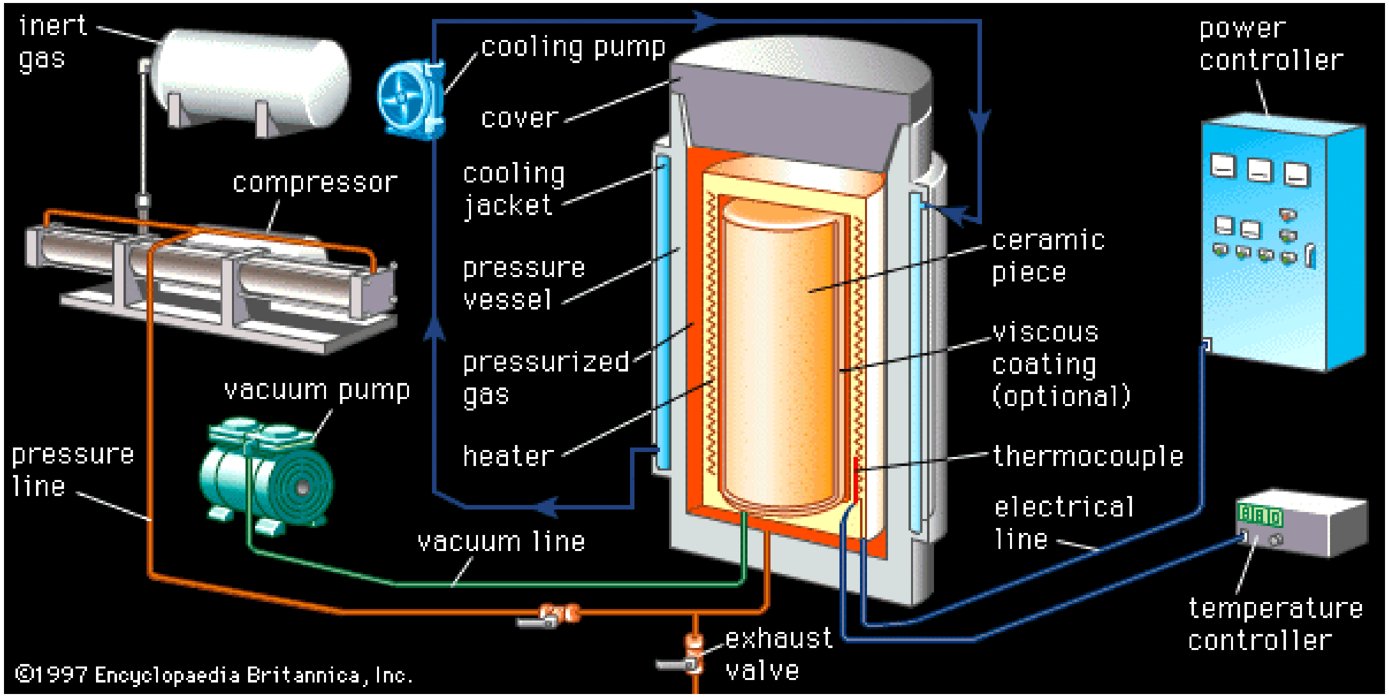




TEM image showing a grain boundary in SiC sintered with rare earth oxides. The latter form a glassy layer along the boundary (Biswas, thesis, 2002)

Hot isostatic pressing (HIP)

Applying an external isostatic pressure will enhance the driving forces of sintering (20-30 times for pressures of 30 to 70 MPa). HIPping will, therefore, shorten the sintering time and increase the final density. Some ceramic material can only be sintered to >95% density if pressure is applied, f.ex. AlN, a ceramic used as high power substrate.



Schematic drawing of a HIP furnace

Hipping of Si_3N_4

sinter mode S: normal HP: hipping	sinter aid appearance of a liquid	% of theoretical density reached	mechanical strenght RT [MPa]	1350 °C [MPa]
Si_3N_4 (HP)	5% MgO	> 98	587	173
Si_3N_4 (S)	5% MgO	90	483	138
Si_3N_4 (HP)	1% MgO	> 99	952	414
Si_3N_4 (S)	$BeSiN_2 + SiO_2$	> 99	560	-
Si_3N_4 (S)	6% Y_2O_3	98	587	414
Si_3N_4 (HP)	13% Y_2O_3	> 99	897	669

(HP): Heissgepresst, (S): gesintert

Hipping of silicon nitride improves the final density and as a consequence the mechanical strength. An improvement of almost 100% is possible.

Reactive sintering

In polyphase ceramics, sintering can be accompanied by chemical reaction. Examples:

Alumina and titaniumnitride based cutting tools

$AlN + TiO_2 = Al_2O_3 + TiN$ (under N_2 atmosphere)

The advantage over direct sintering of the final products is, that TiN, a conducting material, forms a continous network through reactive sintering, allowing machining through electroerosion.

Aluminiumtitanate (low thermal expansion coefficients) for thermal shock application

$Al_2O_3 + TiO_2 = Al_2TiO_2$

Aluminiumtitanate is used as exhaustpipe connectors (Porsche 944)

Sintering recapitulation

Factors influencing solid state sintering:

1. Temperature: densification, T→ sintering rate

2. Green density→final density of sintered body

3. Uniformity of green body density → final density

4. Atmosphere

5. Impurities: - Sintering aids (liquid phase sintering)
   - Suppressing of coarsening
   - Suppressing of anomalous grain growth